Mathematics

Class 1 (English Medium)

Solved Exercises — Sindh Textbook Board

12 lessons — 117 questions

Numbers

Question: Count the things in the pictures and write the number: one mango, two bananas, three apples, four lemons, five pomegranates, six stars, seven candles, eight pencils and nine hens are shown. Write the number of each thing.

Answer: Step 1: Touch and count each thing one by one. Step 2: There is 1 mango, so write 1. Step 3: There are 2 bananas, so write 2. Step 4: Apples are 3, lemons are 4 and pomegranates are 5. Step 5: Stars are 6, candles are 7, pencils are 8 and hens are 9. Final answer: 1, 2, 3, 4, 5, 6, 7, 8, 9. In words: One, Two, Three, Four, Five, Six, Seven, Eight, Nine.

Question: Activity: Count the things in the pictures and write the number: bag, buttons, stars, balloons, pencils, carrots, bananas, birds and mangoes.

Answer: Step 1: Count the things of each group one by one. Step 2: Bag is 1, buttons are 2, stars are 3. Step 3: Balloons are 4, pencils are 5, carrots are 6. Step 4: Bananas are 7, birds are 8 and mangoes are 9. Final answer: bag 1, buttons 2, stars 3, balloons 4, pencils 5, carrots 6, bananas 7, birds 8, mangoes 9.

Question: The concept of zero: Four cages have 3, 2, 1 and 0 parrots. Write the number of parrots in each cage and tell which number is written for the empty cage.

Answer: Step 1: Count the parrots in the first cage: 1, 2, 3 — number 3. Step 2: The second cage has 2 parrots — number 2. Step 3: The third cage has 1 parrot — number 1. Step 4: The fourth cage is completely empty, it has no parrot. Step 5: When there is nothing, we write the number 0 (zero). Final answer: 3, 2, 1, 0. Zero means "nothing".

Question: Activity: Match the plates of apples with the correct number. The plates have apples: 2, 3, 5, 0, 1, 9, 8, 4 (given as example), 6, 7.

Answer: Step 1: Count the apples on each plate. Step 2: Match the plate with 2 apples to number 2. Step 3: In the same way match 3 apples to 3, 5 apples to 5 and the empty plate to 0. Step 4: Match the remaining plates to 1, 9, 8, 6 and 7 (the plate of 4 is given as an example). Final answer: 2—2, 3—3, 5—5, 0—0, 1—1, 9—9, 8—8, 4—4, 6—6, 7—7.

Question: Activity: Count the things in the park picture and write the number: goat, cat, footballs, parrots, ducks, trees, frogs, pigeons, hens and sparrows.

Answer: Step 1: Count every thing in the park carefully. Step 2: Goat is 1 and cat is 1. Step 3: Footballs are 3 and parrots are 4. Step 4: Ducks are 5 and trees are 5. Step 5: Frogs are 6, pigeons are 7, hens are 8 and sparrows are 9. Final answer: goat 1, cat 1, footballs 3, parrots 4, ducks 5, trees 5, frogs 6, pigeons 7, hens 8, sparrows 9.

Forward counting (0 to 9): Fill in the blanks: (i) 0, 1, _____, 3, _____, _____, 6, _____, _____, 9 (ii) 0, _____, 2, _____, _____, 5, _____, _____, 8, _____

Answer: 2, 4, 5, 7, 8, 1, 3, 4, 6, 7, 9

Backward counting (9 to 0): Fill in the blanks: (i) 9, _____, 7, _____, _____, 4, _____, 2, _____, 0 (ii) 9, _____, _____, 6, 5, _____, 3, _____, _____, _____

Answer: 8, 6, 5, 3, 1, 8, 7, 4, 2, 1, 0

Question: Exercise 1, Q1: Count the things in the pictures and write the number: mangoes (4 given as example), birds, ducks, apples, balloons, parrots and cricket balls.

Answer: Step 1: Count the things in each picture one by one. Step 2: Birds are 8, write 8. Step 3: Ducks are 3 and apples are 2. Step 4: Balloons are 5

and parrots are 7. Step 5: Cricket balls are 9. Final answer: mangoes 4 (given), birds 8, ducks 3, apples 2, balloons 5, parrots 7, cricket balls 9.

Question: Exercise 1, Q2: Count the things and match with the correct number: bananas (4 given), mugs, glasses, rabbits, stars, pencils, books, footballs and carrots.

Answer: Step 1: Count the things of each group. Step 2: Mug is 1, match it with 1. Step 3: Glasses are 8 and rabbits are 2. Step 4: Stars are 9, pencils are 5 and books are 3. Step 5: Footballs are 7 and carrots are 6. Final answer: bananas 4 (given), mug 1, glasses 8, rabbits 2, stars 9, pencils 5, books 3, footballs 7, carrots 6.

Exercise 1, Q3: Complete the forward counting rows: (a) 2, _____, _____, 5, _____, 7 (b) 3, _____, 5, _____, _____, 8 (c) 0, _____, 2, _____, _____, 5

Answer: 3, 4, 6, 4, 6, 7, 1, 3, 4

Exercise 1, Q4: Complete the backward counting rows: (a) 9, _____, 7, _____, _____, 4 (b) 8, _____, _____, 5, _____, 3 (c) 5, _____, 3, _____, _____, 0

Answer: 8, 6, 5, 7, 6, 4, 4, 2, 1

Question: Concept of ten: 10 ones make one ten. Tell: 10 ones = ___ ten = ___; and how much are 2, 3, 4, 5, 6, 7, 8, 9 tens?

Answer: Step 1: Count 10 things one by one: 1, 2, 3, 4, 5, 6, 7, 8, 9, 10. Step 2: Put the 10 ones together in one group — this is one ten. Step 3: So 10 ones = 1 ten = 10, and 10 is the first two-digit number. Step 4: Count the tens: 2 tens = 20, 3 tens = 30, 4 tens = 40, 5 tens = 50. Step 5: 6 tens = 60, 7 tens = 70, 8 tens = 80, 9 tens = 90. Final answer: 10 ones = 1 ten = 10; 20, 30, 40, 50, 60, 70, 80, 90.

Question: Numbers 11 to 19: Write in tens and ones: 11, 12, 13, 14, 15, 16, 17, 18, 19 — how many tens and ones are in each?

Answer: Step 1: In each number first separate one ten (10). Step 2: Count the ones that are left. Step 3: 11 = 1 ten + 1 one, 12 = 1 ten + 2 ones, 13 = 1 ten + 3 ones, 14 = 1 ten + 4 ones. Step 4: 15 = 1 ten + 5 ones, 16 = 1 ten + 6 ones, 17 = 1 ten + 7 ones. Step 5: 18 = 1 ten + 8 ones, 19 = 1 ten + 9 ones. Final answer: Every number has 1 ten and the ones are 1, 2, 3, 4, 5, 6, 7, 8, 9 in order.

Question: Activity: Forward and backward counting (bigger numbers): (i) In the row 10 to 19 the missing numbers are ___, ___; 30, 31, 32, ___, 34, 35, ___, 37; 61, 62, ___, 64, ___, 66, ___, 68 (ii) 20, 19, 18, ___, 16, 15, ___, 13, 12; 50, 49, 48, ___, 46, ___, ___, 43; 90, 89, 88, ___, 86, 85, ___, 83

Answer: Step 1: In forward rows add one each time. Step 2: (i) In 10 to 19 the missing numbers are 14, 18; in the second row 33, 36; in the third 63, 65, 67. Step 3: In backward rows take away one each time. Step 4: (ii) In the first row 17, 14; in the second 47, 45, 44; in the third 87, 84. Final answer: (i) 14, 18; 33, 36; 63, 65, 67 (ii) 17, 14; 47, 45, 44; 87, 84.

Question: Exercise 2, Q1: Count the boxes of tens and the ones and write the number: stars (22 given as example: 2 boxes of tens and 2 ones), pencils, footballs and carrots.

Answer: Step 1: First count the full boxes of tens, then the loose ones. Step 2: Pencils: 2 boxes = 20 and 4 ones = 24. Step 3: Footballs: 3 boxes = 30 and 4 ones = 34. Step 4: Carrots: 4 boxes = 40 and 4 ones = 44. Final answer: stars 22 (given), pencils 24, footballs 34, carrots 44.

Question: Exercise 2, Q2: Match the tens-ones with the correct number: 5 tens 6 ones (56 given as example), 7 tens 2 ones, 9 tens 7 ones, 2 tens 4 ones, 4 tens 5 ones.

Answer: Step 1: Count the tens by 10 and add the ones. Step 2: 7 tens = 70 and 2 ones = 72. Step 3: 9 tens = 90 and 7 ones = 97. Step 4: 2 tens 4 ones = 24; 4 tens 5 ones = 45. Final answer: 5 tens 6 ones—56 (given), 7

tens 2 ones—72, 9 tens 7 ones—97, 2 tens 4 ones—24, 4 tens 5 ones—45.

Exercise 2, Q3: Complete the forward counting: (a) 17, 18, _____, 20, 21, _____, _____, 24, _____ (b) 40, 41, _____, 43, _____, _____, 46 (c) 75, _____, 77, _____, _____, 80

Answer: 19, 22, 23, 25, 42, 44, 45, 76, 78, 79

Exercise 2, Q4: Complete the backward counting: (a) 90, _____, 88, _____, 86, _____, 84 (b) 16, 15, _____, 13, _____, 11 (c) 50, _____, 48, _____, 46, _____

Answer: 89, 87, 85, 14, 12, 49, 47, 45

Question: Activity: Hundred chart (1 to 100) — write the missing numbers in each row: the 0s row, 10s, 20s, 30s, 40s, 50s, 60s, 70s, 80s and 90s rows.

Answer: Step 1: Every row of the chart starts from a ten and the numbers increase by one. Step 2: Missing in the 0s row: 4, 6; in the 10s row: 12, 15, 18. Step 3: In the 20s row: 21, 22, 27, 28; in the 30s row: 30, 32, 33, 34. Step 4: In the 40s row: 43, 44, 45, 49; in the 50s row: 52, 53, 56, 58. Step 5: In the 60s row: 61, 63, 65, 68, 69; in the 70s row: 72, 73, 76, 78. Step 6: In the 80s row: 80, 84, 85, 89; in the 90s row: 91, 92, 96, 98. Final answer: 4, 6 | 12, 15, 18 | 21, 22, 27, 28 | 30, 32, 33, 34 | 43, 44, 45, 49 | 52, 53, 56, 58 | 61, 63, 65, 68, 69 | 72, 73, 76, 78 | 80, 84, 85, 89 | 91, 92, 96, 98.

Exercise 3, Q1: Complete the rows of missing numbers: (i) 0, 1, _____, _____, 4, _____, 6 (ii) 9, 10, _____, _____, _____, 14, _____, 16, _____ (iii) 41, 42, _____, _____, 45, _____, _____, 48 (iv) 90, 91, _____, _____, 94, _____, _____, 97 (v) 85, _____, 83, _____, _____, 80, _____, _____, 77 (vi) 50, 49, _____, _____, 46, _____, _____, 43 (vii) 64, 63, _____, _____, _____, 59, _____, 57

Answer: 2, 3, 5, 11, 12, 13, 15, 17, 43, 44, 46, 47, 92, 93, 95, 96, 84, 82,

81, 79, 78, 48, 47, 45, 44, 62, 61, 60, 58

Question: Exercise 3, Q2, Q3 and Q4: (Q2) Match the number words with the correct figures: Six, Zero, Nine, Two, Five, Three (given), Four, Eight, One, Ten, Seven. (Q3) Write the words in figures: Nine (given 9), Five, Three, Seven, Four, Eight. (Q4) Write the figures in words: 10 (given Ten), 2, 7, 6, 8, 1.

Answer: Step 1: First read each word aloud and recall its counting. Step 2: Q2: Six = 6, Zero = 0, Nine = 9, Two = 2, Five = 5. Step 3: Three = 3 (given), Four = 4, Eight = 8, One = 1, Ten = 10, Seven = 7. Step 4: Q3: Five = 5, Three = 3, Seven = 7, Four = 4, Eight = 8. Step 5: Q4: 2 = Two, 7 = Seven, 6 = Six, 8 = Eight, 1 = One. Final answer: Q2: 6, 0, 9, 2, 5, 3, 4, 8, 1, 10, 7; Q3: 5, 3, 7, 4, 8; Q4: Two, Seven, Six, Eight, One.

Question: Place value: (i) Tell the place value of 2 and 6 in 26 and write 26 in expanded form. (ii) Write 65 in expanded form. (iii) Tell the place value of 7 in 74 and of 9 in 89. (iv) Write the abacus numbers: 5 tens 3 ones (given), 3 tens 2 ones, 2 tens 4 ones; and 5 tens 6 ones (given), 7 tens 2 ones, 8 tens 3 ones, 1 ten 6 ones.

Answer: Step 1: The first (left) digit is in the tens place; its value = digit × 10. Step 2: (i) In 26, 2 is in the tens place, place value 20; 6 is in the ones place, place value 6; 26 = 20 + 6. Step 3: (ii) 65 = 60 + 5. Step 4: (iii) In 74 the place value of 7 is 7 tens = 70; in 89 the place value of 9 is 9 ones = 9. Step 5: (iv) Abacus numbers: 53 (given), 32, 24; and 56 (given), 72, 83, 16. Final answer: (i) 20 and 6; 26 = 20 + 6 (ii) 65 = 60 + 5 (iii) 70; 9 (iv) 53, 32, 24; 56, 72, 83, 16.

Question: Exercise 4, Q1: Write the place value of the digit in the blue box: 4 in 74 (given), 3 in 35, 4 in 42, 7 in 87.

Answer: Step 1: See whether the boxed digit is in the tens place or the ones place. Step 2: In 74, 4 is in the ones place = 4 ones = 4 (given). Step

3: In 35, 3 is in the tens place = 3 tens = 30. Step 4: In 42, 4 = 4 tens = 40; in 87, 7 is in the ones place = 7. Final answer: 4 = 4 (given), 3 = 30, 4 = 40, 7 = 7.

Question: Exercise 4, Q2 and Q3: (Q2) Write the place value of the encircled digit: 5 in 58 (given), 9 in 89, 8 in 28, 1 in 14. (Q3) Write the abacus numbers: 5 tens 3 ones (given), 2 tens 1 one, 2 tens 5 ones, 6 tens 1 one.

Answer: Step 1: Identify the place of the encircled digit. Step 2: Q2: In 58, 5 = 5 tens = 50 (given); in 89, 9 = 9 ones = 9. Step 3: In 28, 8 = 8; in 14, 1 = 1 ten = 10. Step 4: Q3: Count the tens by 10 and add the ones: 21, 25, 61. Final answer: Q2: 50 (given), 9, 8, 10; Q3: 53 (given), 21, 25, 61.

Question: Exercise 5, Q1: Compare with the given number and tell whether the other number is less, equal or more: (i) Given 26: 11, 26, 39 (ii) Given 44: 57, 28, 44 (iii) Given 72: 72, 89, 39 (iv) Given 86: 92, 74, 86.

Answer: Step 1: Compare the tens of both numbers; if the tens are equal, compare the ones. Step 2: (i) With 26: 11 is less (1 ten < 2), 26 is equal, 39 is more. Step 3: (ii) With 44: 57 is more, 28 is less, 44 is equal. Step 4: (iii) With 72: 72 is equal, 89 is more, 39 is less. Step 5: (iv) With 86: 92 is more, 74 is less, 86 is equal. Final answer: (i) less, equal, more (ii) more, less, equal (iii) equal, more, less (iv) more, less, equal.

Question: Exercise 5, Q2: Compare with the given number and colour the greater number blue, the smaller red and the equal yellow: (i) Given 15: 32, 12, 15 (ii) Given 58: 72, 58, 34 (iii) Given 63: 63, 24, 74 (iv) Given 90: 82, 90, 97.

Answer: Step 1: Compare each number with the given number. Step 2: (i) With 15: 32 is greater (blue), 12 is smaller (red), 15 is equal (yellow). Step 3: (ii) With 58: 72 is greater (blue), 58 is equal (yellow), 34 is smaller (red). Step 4: (iii) With 63: 63 is equal (yellow), 24 is smaller (red), 74 is greater

(blue). Step 5: (iv) With 90: 82 is smaller (red), 90 is equal (yellow), 97 is greater (blue). Final answer: (i) blue, red, yellow (ii) blue, yellow, red (iii) yellow, red, blue (iv) red, yellow, blue.

Question: Exercise 5, Q3 and Q4: (Q3) Write in increasing order (smallest to greatest): 27, 46, 54, 11 (given solved); 95, 13, 57, 82; 9, 66, 39, 26; 88, 22, 77, 49. (Q4) Write in decreasing order (greatest to smallest): 31, 15, 55, 73 (given solved); 14, 23, 59, 69; 80, 90, 70, 50; 11, 77, 23, 73.

Answer: Step 1: For increasing order first find the smallest number, then the next bigger, and so on till the end. Step 2: Q3: The tens of 95, 13, 57, 82 are 9, 1, 5, 8 — order: 13, 57, 82, 95. Step 3: 9, 66, 39, 26 — order: 9, 26, 39, 66; and 88, 22, 77, 49 — order: 22, 49, 77, 88. Step 4: For decreasing order start from the greatest: 14, 23, 59, 69 — order: 69, 59, 23, 14. Step 5: 80, 90, 70, 50 — order: 90, 80, 70, 50; and 11, 77, 23, 73 — order: 77, 73, 23, 11. Final answer: Q3: 11, 27, 46, 54 (given); 13, 57, 82, 95; 9, 26, 39, 66; 22, 49, 77, 88. Q4: 73, 55, 31, 15 (given); 69, 59, 23, 14; 90, 80, 70, 50; 77, 73, 23, 11.

Question: Exercise 5, Q5: Write the Before, After and Between numbers: Before: 34, 56, 98, 88, 92, 1 (25, 26 given); After: 73, 19, 44, 63, 84, 33 (51, 52 given); Between: 36-38, 90-92, 87-89, 53-55, 0-2, 11-13 (42, 43, 44 given).

Answer: Step 1: "Before" number = 1 less than the given number; "After" = 1 more; "Between" is exactly in the middle of the two. Step 2: Before: 34 → 33, 56 → 55, 98 → 97. Step 3: Before: 88 → 87, 92 → 91, 1 → 0. Step 4: After: 73 → 74, 19 → 20, 44 → 45, 63 → 64, 84 → 85, 33 → 34. Step 5: Between: between 36 and 38 is 37; in 90-92 is 91; in 87-89 is 88; in 53-55 is 54; in 0-2 is 1; in 11-13 is 12. Final answer: Before: 33, 55, 97, 87, 91, 0; After: 74, 20, 45, 64, 85, 34; Between: 37, 91, 88, 54, 1, 12.

Question: Exercise 5, Q6 and Q7: (Q6) Complete the pairs: 57, ___; ___, 69;

29, ___; ___, 34; 77, ___. (Q7) Complete the triples: 62, ___, 64; 36, 37, ___; ___, 55, 56; 19, ___, 21; 73, ___, 75; ___, 88, 89.

Answer: Step 1: In Q6 the second number of each pair is 1 more or 1 less than the first — decide by looking at the given number. Step 2: After 57 is 58; before 69 is 68; after 29 is 30; before 34 is 33; after 77 is 78. Step 3: In Q7 fill the middle or corner number by the same rule. Step 4: 62, 63, 64; 36, 37, 38; 54, 55, 56; 19, 20, 21; 73, 74, 75; 87, 88, 89. Final answer: Q6: 58; 68; 30; 33; 78. Q7: 63; 38; 54; 20; 74; 87.

Question: Exercise 6, Q1: Match the ordinal numbers with their words: 1st, 2nd, 3rd, 4th, 5th (Fifth given), 6th, 7th, 8th, 9th, 10th.

Answer: Step 1: An ordinal number tells the position of a thing. Step 2: 1st = First, 2nd = Second, 3rd = Third, 4th = Fourth. Step 3: 5th = Fifth (given), 6th = Sixth, 7th = Seventh. Step 4: 8th = Eighth, 9th = Ninth, 10th = Tenth. Final answer: 1st—First, 2nd—Second, 3rd—Third, 4th—Fourth, 5th—Fifth, 6th—Sixth, 7th—Seventh, 8th—Eighth, 9th—Ninth, 10th—Tenth.

Question: Exercise 6, Q2 and Q3: (Q2) Write in words: 5th (Fifth given), 10th, 8th, 2nd, 1st, 6th. (Q3) Write in figures: Fourth (4th given), Seventh, Ninth, Third, Fifth.

Answer: Step 1: First recall the order words: First, Second, Third, Fourth, Fifth. Step 2: Q2: 10th = Tenth, 8th = Eighth, 2nd = Second. Step 3: 1st = First, 6th = Sixth. Step 4: Q3: Seventh = 7th, Ninth = 9th, Third = 3rd, Fifth = 5th. Final answer: Q2: Fifth (given), Tenth, Eighth, Second, First, Sixth; Q3: 4th (given), 7th, 9th, 3rd, 5th.

Question: Exercise 6, Q4: Write the ordinal position of the days of the week: Monday (1st given), Tuesday, Wednesday, Thursday, Friday, Saturday, Sunday.

Answer: Step 1: The week starts from Monday; count the days in order.

Step 2: Monday is the first day = 1st (given); Tuesday = 2nd; Wednesday = 3rd. Step 3: Thursday = 4th; Friday = 5th. Step 4: Saturday = 6th; Sunday is the seventh and last day = 7th. Final answer: Monday 1st, Tuesday 2nd, Wednesday 3rd, Thursday 4th, Friday 5th, Saturday 6th, Sunday 7th.

Question: Activity: In the race of ten cars, the position of each car is written on its flag: red 1st, light-purple 2nd, magenta 3rd, dark-blue 4th, green 5th, light-blue 6th, purple 7th, blue 8th, light-red 9th, grey 10th. In the table write the position of the green, blue, light-purple and magenta cars in figures and in words.

Answer: Step 1: Read the ordinal number written on the flag of each car. Step 2: The green car's flag says 5th — Fifth. Step 3: The blue car is 8th — Eighth. Step 4: Light-purple is 2nd — Second and magenta is 3rd — Third. Final answer: Green 5th Fifth, Blue 8th Eighth, Light-purple 2nd Second, Magenta 3rd Third. (Note: the "3rd" printed in the table's example row does not match the red car's flag 1st — the flags show the correct positions.)

Question: Activity: In the motorbike race some positions are missing. Given flags: green 1st, purple 3rd, magenta 5th, light-purple 7th, red 10th. Fill in the missing positions: orange, lime-green, brown, yellow and pink motorbikes.

Answer: Step 1: Put the motorbikes in race order: from 1st to 10th. Step 2: Find the empty numbers between the given flags: between 1st and 3rd is 2nd — that is the orange bike. Step 3: Between 3rd and 5th is 4th — lime-green; between 5th and 7th is 6th — brown. Step 4: After 7th comes 8th — yellow; then 9th — pink; at the end 10th red (given). Final answer: orange 2nd, lime-green 4th, brown 6th, yellow 8th, pink 9th.

Number Operations

Introduction to addition: (i) 4 apples + 3 apples = how many apples in all? (ii) 3 balloons + 5 balloons = _____ balloons. (iii) 5 bananas + 6 bananas = _____ and 6 stars + 8 stars = _____. Tell the meaning of "+" and "=".

Answer: 8 balloons, 11, 14

Question: Addition facts (up to 20): Complete: 1 + 1, 2 + 1, 3 + 1, 4 + 1, 5 + 1; 1 + 2, 2 + 2, 3 + 2, 4 + 2, 5 + 2; 1 + 3, 2 + 3, 3 + 3, 4 + 3, 5 + 3; 6 + 4, 7 + 4, 8 + 4, 9 + 4; 6 + 5, 7 + 5, 8 + 5, 9 + 5; 6 + 6, 7 + 6, 8 + 6, 9 + 6; 7 + 7, 8 + 7, 9 + 7, 10 + 7; 7 + 8, 8 + 8, 9 + 8, 10 + 8; 7 + 9, 8 + 9, 9 + 9, 10 + 9; 7 + 10, 8 + 10, 9 + 10, 10 + 10.

Answer: Step 1: Count forward from the bigger number to add (for example for 4 + 3 count 3 forward from 4: 5, 6, 7). Step 2: Of 1: 2, 3, 4, 5, 6; of 2: 3, 4, 5, 6, 7; of 3: 4, 5, 6, 7, 8. Step 3: Of 4: 10, 11, 12, 13; of 5: 11, 12, 13, 14; of 6: 12, 13, 14, 15. Step 4: Of 7: 14, 15, 16, 17; of 8: 15, 16, 17, 18. Step 5: Of 9: 16, 17, 18, 19; of 10: 17, 18, 19, 20. Final answer: 2, 3, 4, 5, 6 | 3, 4, 5, 6, 7 | 4, 5, 6, 7, 8 | 10, 11, 12, 13 | 11, 12, 13, 14 | 12, 13, 14, 15 | 14, 15, 16, 17 | 15, 16, 17, 18 | 16, 17, 18, 19 | 17, 18, 19, 20.

Question: Introduction to subtraction: (i) There were 6 eggs, 2 broke — how many are left? (ii) There were 5 parrots, 2 flew away — how many are left? (iii) There were 8 lemons in all, 2 were taken — how many are left? (iv) Write the related subtraction fact: from 8 + 2 = 10; from 9 + 6 = 15; from 8 + 9 = 17; from 12 + 8 = 20. (v) Complete: 6 + 3, 14 + 6, 17 + 2; 9 − 6, 20 − 14, 19 − 2.

Answer: Step 1: Subtraction means taking some away from the total and counting what is left; its sign is "−". Step 2: (i) 6 − 2 = 4 eggs left; (ii) 5 −

2 = 3 parrots left; (iii) 8 − 2 = 6 lemons left. Step 3: (iv) Related subtraction facts: 10 − 8 = 2; 15 − 6 = 9; 17 − 9 = 8; 20 − 8 = 12. Step 4: (v) 6 + 3 = 9; 14 + 6 = 20; 17 + 2 = 19. Step 5: 9 − 6 = 3; 20 − 14 = 6; 19 − 2 = 17. Final answer: (i) 4 (ii) 3 (iii) 6 (iv) 10 − 8 = 2; 15 − 6 = 9; 17 − 9 = 8; 20 − 8 = 12 (v) 9, 20, 19; 3, 6, 17.

Question: Exercise 1, Q1: Add and subtract from the pictures: (i) 3 pomegranates + 2 pomegranates (ii) 5 books + 4 books (iii) 6 apples − 2 apples (iv) 6 pencils − 3 pencils.

Answer: Step 1: First count the things in the picture, then add or subtract by looking at the sign. Step 2: (i) Join the pomegranates: 3 + 2 = 5. Step 3: (ii) Join the books: 5 + 4 = 9. Step 4: (iii) Take away 2 apples: 6 − 2 = 4. Step 5: (iv) Take away 3 pencils: 6 − 3 = 3. Final answer: (i) 5 (ii) 9 (iii) 4 (iv) 3.

Question: Exercise 1, Q2: Match each sentence with its correct answer: 5 + 7, 8 + 6, 9 + 8, 8 − 3, 10 − 4 (6 given), 18 − 2. Answer set: 12, 14, 17, 5, 6, 16.

Answer: Step 1: Solve each sentence separately, then match it with the answer. Step 2: 5 + 7: count 7 forward from 5: 6, 7, 8, 9, 10, 11, 12 — answer 12. Step 3: 8 + 6 = 14; 9 + 8 = 17. Step 4: 8 − 3 = 5; 10 − 4 = 6 (given); 18 − 2 = 16. Final answer: 5 + 7—12, 8 + 6—14, 9 + 8—17, 8 − 3—5, 10 − 4—6, 18 − 2—16.

Question: Exercise 1, Q3: Complete: 2 + 5 (7 given), 9 + 8, 16 + 4, 7 + ___ = 13, 11 + ___ = 20; 17 − 5 (12 given), 15 − 8, 20 − 16, 13 − ___ = 6, 20 − ___ = 11.

Answer: Step 1: Solve the straight sentences first. Step 2: 9 + 8 = 17; 16 + 4 = 20. Step 3: For a missing addend subtract the smaller number from the bigger: 13 − 7 = 6, so 7 + 6 = 13; 20 − 11 = 9, so 11 + 9 = 20. Step 4: 15 − 8 = 7; 20 − 16 = 4. Step 5: For a missing subtraction number: 13

− 6 = 7, so 13 − 7 = 6; 20 − 11 = 9, so 20 − 9 = 11. Final answer: 7 (given), 17, 20, 6, 9; 12 (given), 7, 4, 7, 9.

Question: Exercise 1, Q4: Match the equal addition sentences (changing the order in addition does not change the answer): 5 + 7, 13 + 4, 17 + 3 (given), 8 + 6, 15 + 4 — pairs: 7 + 5, 4 + 13, 3 + 17, 6 + 8, 4 + 15.

Answer: Step 1: Look at both numbers of each sentence and find the same numbers in reverse order. Step 2: The pair of 5 + 7 is 7 + 5 — both answers are 12. Step 3: The pair of 13 + 4 is 4 + 13 — answer 17; the pair of 17 + 3 is 3 + 17 (given) — answer 20. Step 4: The pair of 8 + 6 is 6 + 8 — answer 14; the pair of 15 + 4 is 4 + 15 — answer 19. Final answer: 5 + 7—7 + 5, 13 + 4—4 + 13, 17 + 3—3 + 17, 8 + 6—6 + 8, 15 + 4—4 + 15.

Question: Exercise 2, Q1: Comparing "more": (i) Cups 4 and 6 (given solved) (ii) Bottles 3 and 7 (iii) Roses 8 and 5 — tell how much more the bigger group is than the smaller.

Answer: Step 1: Count the things of both groups. Step 2: To find the difference, subtract the smaller number from the bigger. Step 3: (i) 6 − 4 = 2 — 6 is more than 4 by 2 (given). Step 4: (ii) 7 − 3 = 4 — 7 is more than 3 by 4. Step 5: (iii) 8 − 5 = 3 — 8 is more than 5 by 3. Final answer: (i) more by 2 (ii) more by 4 (iii) more by 3.

Question: Exercise 2, Q2: Comparing "less": (i) Fish 4 and 7 (given solved) (ii) Balloons 5 and 3 (iii) Lemons 5 and 10 — tell how much less the smaller group is than the bigger.

Answer: Step 1: Count the things of both groups. Step 2: Difference = bigger number − smaller number. Step 3: (i) 7 − 4 = 3 — 4 is less than 7 by 3 (given). Step 4: (ii) 5 − 3 = 2 — 3 is less than 5 by 2. Step 5: (iii) 10 − 5 = 5 — 5 is less than 10 by 5. Final answer: (i) less by 3 (ii) less by 2 (iii) less by 5.

Question: Activity: Mental and vertical addition/subtraction: (i) 6 + 2 (8

given), 5 + 0, 5 + 7 (ii) Three numbers: 6 + 4 + 5 (given), 7 + 4 + 2, 5 + 6 + 3 (iii) Vertical: 7 + 5, 9 + 8, 4 + 5, 3 + 9; 4 + 5 + 3 (12 given), 6 + 4 + 8.

Answer: Step 1: (i) Count forward in your mind: 5 + 0 = 5 (adding zero keeps the number the same); 5 + 7: count 7 forward from 5 = 12. Step 2: (ii) In three numbers add the first two, then the third: 6 + 4 = 10, 10 + 5 = 15 (given); 7 + 4 = 11, 11 + 2 = 13; 5 + 6 = 11, 11 + 3 = 14. Step 3: (iii) In vertical form write the ones under each other and add: 7 + 5 = 12; 9 + 8 = 17; 4 + 5 = 9; 3 + 9 = 12. Step 4: 4 + 5 + 3: 4 + 5 = 9, 9 + 3 = 12 (given); 6 + 4 + 8: 6 + 4 = 10, 10 + 8 = 18. Final answer: (i) 8 (given), 5, 12 (ii) 15 (given), 13, 14 (iii) 12, 17, 9, 12; 12 (given), 18.

Question: Exercise 3, Q1: Add and subtract mentally: 5 + 4 (9 given), 8 + 7, 6 + 7, 9 + 0, 3 + 5; 9 − 3 (6 given), 8 − 5, 5 − 2, 7 − 6, 4 − 0.

Answer: Step 1: For addition count forward from the bigger number; for subtraction count backward. Step 2: 8 + 7: count 7 forward from 8: 9, 10, 11, 12, 13, 14, 15 = 15; 6 + 7 = 13. Step 3: 9 + 0 = 9; 3 + 5 = 8. Step 4: 8 − 5: count 5 back from 8: 7, 6, 5, 4, 3 = 3; 5 − 2 = 3. Step 5: 7 − 6 = 1; 4 − 0 = 4 (subtracting zero keeps the number the same). Final answer: 9 (given), 15, 13, 9, 8; 6 (given), 3, 3, 1, 4.

Question: Exercise 3, Q2: Add and subtract mentally (with tens): 35 + 20 (55 given), 17 + 40, 64 + 30, 54 + 10, 27 + 50; 65 − 20 (45 given), 92 − 50, 78 − 40, 59 − 30, 48 − 30.

Answer: Step 1: When the second number is only tens, the ones stay the same and only the tens change. Step 2: 17 + 40: ones stay 7, tens 1 + 4 = 5 — answer 57; 64 + 30 = 94; 54 + 10 = 64; 27 + 50 = 77. Step 3: 92 − 50: tens 9 − 5 = 4, ones 2 — answer 42. Step 4: 78 − 40 = 38; 59 − 30 = 29; 48 − 30 = 18. Final answer: 55 (given), 57, 94, 64, 77; 45 (given), 42, 38, 29, 18.

Question: Exercise 3, Q3: Add and subtract the following (vertical): 5 + 4, 6 + 2, 36 + 20, 47 + 30, 59 + 40, 65 + 30, 28 + 30, 81 + 50, 73 + 40; 9 − 4, 8 − 3, 23 − 10, 85 − 30, 77 − 40, 45 − 20, 82 − 20, 45 − 30, 29 − 10.

Answer: Step 1: In the vertical method first add/subtract the ones, then the tens. Step 2: Addition: 5 + 4 = 9; 6 + 2 = 8; 36 + 20: ones 6 + 0 = 6, tens 3 + 2 = 5 — 56. Step 3: 47 + 30 = 77; 59 + 40 = 99; 65 + 30 = 95; 28 + 30 = 58. Step 4: 81 + 50: ones 1 + 0 = 1, tens 8 + 5 = 13 — 131; 73 + 40 = 113. Step 5: Subtraction: 9 − 4 = 5; 8 − 3 = 5; 23 − 10 = 13; 85 − 30 = 55; 77 − 40 = 37; 45 − 20 = 25; 82 − 20 = 62; 45 − 30 = 15; 29 − 10 = 19. Final answer: 9, 8, 56, 77, 99, 95, 58, 131, 113; 5, 5, 13, 55, 37, 25, 62, 15, 19.

Question: Exercise 4, Q1: Add and subtract: 47 + 2, 15 + 3, 92 + 6, 32 + 54, 51 + 43; 38 − 5, 47 − 7, 88 − 5, 75 − 42, 96 − 43.

Answer: Step 1: First solve the ones, then the tens. Step 2: 47 + 2: ones 7 + 2 = 9, tens stay 4 — 49; 15 + 3 = 18; 92 + 6 = 98. Step 3: 32 + 54: ones 2 + 4 = 6, tens 3 + 5 = 8 — 86; 51 + 43 = 94. Step 4: 38 − 5: ones 8 − 5 = 3 — 33; 47 − 7 = 40; 88 − 5 = 83. Step 5: 75 − 42: ones 5 − 2 = 3, tens 7 − 4 = 3 — 33; 96 − 43 = 53. Final answer: 49, 18, 98, 86, 94; 33, 40, 83, 33, 53.

Question: Exercise 4, Q3 (the book's numbering jumps from 1 to 3): Add and subtract vertically: 35 + 3, 62 + 7, 93 + 4, 24 + 3, 24 + 63, 66 + 23, 12 + 85, 48 + 30; 48 − 4, 99 − 8, 56 − 3, 79 − 6, 28 − 15, 43 − 21, 52 − 41, 68 − 35.

Answer: Step 1: Write the numbers with ones under ones and tens under tens. Step 2: Addition (one-digit): 35 + 3: 5 + 3 = 8 — 38; 62 + 7 = 69; 93 + 4 = 97; 24 + 3 = 27. Step 3: Addition (two-digit): 24 + 63: ones 4 + 3 = 7, tens 2 + 6 = 8 — 87; 66 + 23 = 89; 12 + 85 = 97; 48 + 30 = 78. Step

4: Subtraction (one-digit): 48 − 4 = 44; 99 − 8 = 91; 56 − 3 = 53; 79 − 6 = 73. Step 5: Subtraction (two-digit): 28 − 15: 8 − 5 = 3, 2 − 1 = 1 — 13; 43 − 21 = 22; 52 − 41 = 11; 68 − 35 = 33. Final answer: 38, 69, 97, 27, 87, 89, 97, 78; 44, 91, 53, 73, 13, 22, 11, 33.

Question: Word problems (examples): (1) Hina has 20 rupees, her friend gave her 15 rupees — how many in all? (2) Moeen bought 15 kg flour, his son bought 5 kg more — total flour? (3) There are 29 students in the class, 4 are absent — how many are present? (4) The sum of two cards is 54, one card shows 34 — what is on the other card?

Answer: Step 1: Find both numbers and the operation (add or subtract) from the problem. Step 2: (1) Join both amounts: 20 + 15 — ones 0 + 5 = 5, tens 2 + 1 = 3 — 35 rupees. Step 3: (2) 15 + 5: 5 + 5 = 10 (write 0, carry 1 ten), tens 1 + 1 = 2 — 20 kg. Step 4: (3) Subtract the absent: 29 − 4: 9 − 4 = 5 — 25 students present. Step 5: (4) Missing number = total − known: 54 − 34: 4 − 4 = 0, 5 − 3 = 2 — 20. Final answer: (1) 35 rupees (2) 20 kg (3) 25 students (4) 20.

Question: Exercise 5, Q1: The number of girls in a class is 16 and the number of boys in the same class is 23. The total number of students in the class = ___.

Answer: Step 1: To find the total, add the girls and boys: 16 + 23. Step 2: Add the ones: 6 + 3 = 9. Step 3: Add the tens: 1 + 2 = 3. Step 4: Write both together: 3 tens 9 ones = 39. Final answer: There are 39 students in all.

Question: Exercise 5, Q2: Abdul Rafay has 36 rupees and Muhammad Taha has 52 rupees. How much money do they have in all?

Answer: Step 1: Add both amounts: 36 + 52. Step 2: Ones: 6 + 2 = 8. Step 3: Tens: 3 + 5 = 8. Step 4: Result: 8 tens 8 ones = 88. Final answer: They have 88 rupees in all.

Question: Exercise 5, Q3: Faiza bought 60 grams of almonds and Elsa bought 30 grams of almonds. What is the total weight of their almonds?

Answer: Step 1: Add both weights: 60 + 30. Step 2: Ones: 0 + 0 = 0. Step 3: Tens: 6 + 3 = 9. Step 4: Result: 9 tens 0 ones = 90. Final answer: The total weight is 90 grams.

Question: Exercise 5, Q4: A shopkeeper has 46 eggs. He sold 23 eggs. How many eggs are left?

Answer: Step 1: Subtract the sold eggs from the total: 46 − 23. Step 2: Ones: 6 − 3 = 3. Step 3: Tens: 4 − 2 = 2. Step 4: Result: 2 tens 3 ones = 23. Final answer: 23 eggs are left.

Question: Exercise 5, Q5 and Q6: (5) The total price of two toys is 95 rupees. The price of one toy is 40 rupees; tell the price of the other toy. (6) 26 parrots were sitting on a tree, 12 parrots flew away. Tell the number of parrots left on the tree.

Answer: Step 1: (5) Missing price = total price − known price: 95 − 40. Step 2: Ones: 5 − 0 = 5; tens: 9 − 4 = 5 — the other toy is 55 rupees. Step 3: (6) Subtract the parrots that flew away: 26 − 12. Step 4: Ones: 6 − 2 = 4; tens: 2 − 1 = 1 — 14 parrots left. Final answer: (5) 55 rupees (6) 14 parrots.

Question: Rounding and estimation: (i) Round to the nearest ten: 23, 42, 51, 64, 83, 37, 76, 45, 39, 8. (ii) Estimate: 46 + 21; and estimate the difference: 82 − 26.

Answer: Step 1: Rounding rule: if the ones are less than 5, keep the tens the same (round down); if the ones are 5 or more, increase one ten (round up). Step 2: Round down: 23 → 20 (ones 3 < 5), 42 → 40, 51 → 50, 64 → 60, 83 → 80. Step 3: Round up: 37 → 40, 76 → 80, 45 → 50, 39 → 40, 8 → 10. Step 4: (ii) For estimation first round both numbers: 46 → 50 and 21 → 20; estimated sum = 50 + 20 = 70. Step 5: Difference: 82 → 80 and 26 →

30; estimated difference = 80 − 30 = 50. Final answer: (i) 20, 40, 50, 60, 80, 40, 80, 50, 40, 10 (ii) estimated sum 70; estimated difference 50.

Question: Exercise 6, Q1: Match each number with its rounded (nearest ten) form: 72, 86, 17, 28 (30 given), 45, 62 — answers: 70, 90, 20, 30, 50, 60.

Answer: Step 1: Look at the ones of each number: less than 5 = down, 5 or more = up. Step 2: The ones of 72 are 2 < 5 — match with 70. Step 3: The ones of 86 are 6 — with 90; the ones of 17 are 7 — with 20. Step 4: 28 = 30 (given); the ones of 45 are 5 — with 50; the ones of 62 are 2 — with 60. Final answer: 72—70, 86—90, 17—20, 28—30, 45—50, 62—60.

Question: Exercise 6, Q2: Round and estimate: 18 + 58, 32 + 49, 23 + 56, 27 + 61, 64 + 15; 46 − 21, 66 − 53, 84 − 17, 55 − 39, 78 − 22.

Answer: Step 1: First round each number to the nearest ten, then add/subtract. Step 2: 18 → 20, 58 → 60: 20 + 60 = 80; 32 → 30, 49 → 50: 30 + 50 = 80. Step 3: 23 → 20, 56 → 60: 80; 27 → 30, 61 → 60: 90; 64 → 60, 15 → 20: 80. Step 4: Subtraction: 46 → 50, 21 → 20: 50 − 20 = 30; 66 → 70, 53 → 50: 70 − 50 = 20. Step 5: 84 → 80, 17 → 20: 60; 55 → 60, 39 → 40: 20; 78 → 80, 22 → 20: 60. Final answer: 80, 80, 80, 90, 80; 30, 20, 60, 20, 60.

Counting in 2s, 5s and 10s (skip counting): (i) In 2s: 6, _____, 10, _____, 14, _____, 18, _____, 22 (ii) In 5s: 5, _____, 15, _____, _____, 30, _____, _____, 45 (iii) In 10s: 10, _____, 30, _____, _____, 60, _____, 80

Answer: 8, 12, 16, 20, 10, 20, 25, 35, 40, 20, 40, 50, 70

Exercise 7, Q1, Q2 and Q3: Complete the number lines: (Q1) In 2s: 2, 4, _____, 8, _____, 12, _____, 16, _____, 20 (Q2) In 5s: 5, _____, 15, _____, 25, _____, 35, _____, 45 (Q3) In 10s: 10, 20, _____, 40, _____, _____, 70, _____, 90

Answer: 6, 10, 14, 18, 10, 20, 30, 40, 30, 50, 60, 80

Question: Exercise 7, Q4 and Q5: (Q4) In the hundred grid fill the missing squares counting in 2s. (Q5) Fill the missing squares counting in 5s.

Answer: Step 1: In Q4 all the numbers counting in 2s are even numbers: starting from 2, 4, 6, every second number. Step 2: Write row by row: 2, 4, 6, 8, 10; 12, 14, 16, 18, 20; and the same pattern onward up to 98. Step 3: In Q5 the numbers counting in 5s end in 5 or 0. Step 4: They are: 5, 10, 15, 20, 25, 30, 35, 40, 45, 50; 55, 60, 65, 70, 75, 80, 85, 90, 95. Final answer: Q4: all even numbers from 2 to 98; Q5: multiples of 5 from 5 to 95.

Exercise 7, Q6, Q7 and Q8: Complete the rows: (Q6) 8, 10, _____, _____, 16, _____, 20, _____, 24 (Q7) 15, 20, _____, _____, 35, _____, 45, _____, 55 (Q8) 20, 30, _____, 50, _____, 70, _____, 90

Answer: 12, 14, 18, 22, 25, 30, 40, 50, 40, 60, 80

Question: Tables: Write the tables of 2, 5 and 10 (from 1 time to 10 times).

Answer: Step 1: Table of 2 = counting in 2s: 2, 4, 6, 8, 10, 12, 14, 16, 18, 20. Step 2: Read: 1 time 2 = 2, 2 times 2 = 4, 3 times 2 = 6, and so on up to 10 times 2 = 20. Step 3: Table of 5 = counting in 5s: 5, 10, 15, 20, 25, 30, 35, 40, 45, 50. Step 4: Table of 10 = counting in 10s: 10, 20, 30, 40, 50, 60, 70, 80, 90, 100. Step 5: Remember: 10 tens = one hundred = 100. Final answer: Table of 2: 2, 4, 6, 8, 10, 12, 14, 16, 18, 20; table of 5: 5, 10, 15, 20, 25, 30, 35, 40, 45, 50; table of 10: 10, 20, 30, 40, 50, 60, 70, 80, 90, 100.

Question: Exercise 8: (Q1) Match: 4 twos (8 given), 5 twos, 6 twos, 7 twos, 8 twos. (Q2) 3 fives (15 given), 4 fives, 5 fives, 6 fives, 7 fives. (Q3) 2 tens (20 given), 3 tens, 4 tens, 5 tens, 6 tens.

Answer: Step 1: Use the table counting for each one. Step 2: Q1: 5 twos =

10; 6 twos = 12; 7 twos = 14; 8 twos = 16. Step 3: Q2: 4 fives = 20; 5 fives = 25; 6 fives = 30; 7 fives = 35. Step 4: Q3: 3 tens = 30; 4 tens = 40; 5 tens = 50; 6 tens = 60. Final answer: Q1: 8 (given), 10, 12, 14, 16; Q2: 15 (given), 20, 25, 30, 35; Q3: 20 (given), 30, 40, 50, 60.

Question: Multiplication = repeated addition: (i) Total wheels of 4 tricycles (3 wheels each)? (ii) 3 groups of eggs, 2 eggs in each group. (iii) 3 groups of tennis balls, 4 balls in each. (iv) 4 rows of oranges, 5 in each row. (v) 5 rows of apples, 6 in each row. (vi) Match the repeated addition with multiplication: 7 + 7 + 7 (given), 8 + 8 + 8 + 8, 2 + 2 + 2 + 2 + 2 + 2, 4 + 4 + 4 + 4 + 4, 5 + 5 + 5.

Answer: Step 1: When the same number is added again and again, it is multiplication; "×" is the sign of multiplication. Step 2: (i) 3 + 3 + 3 + 3 = 12; read 4 × 3 = 12 wheels. Step 3: (ii) 2 + 2 + 2 = 6; 3 × 2 = 6. Step 4: (iii) 4 + 4 + 4 = 12; 3 × 4 = 12; (iv) 5 + 5 + 5 + 5 = 20; 4 × 5 = 20; (v) 6 + 6 + 6 + 6 + 6 = 30; 5 × 6 = 30. Step 5: (vi) 7 + 7 + 7 = 3 × 7 (given); 8 + 8 + 8 + 8 = 4 × 8; 2 six times = 6 × 2; 4 five times = 5 × 4; 5 + 5 + 5 = 3 × 5. Final answer: (i) 12 (ii) 6 (iii) 12 (iv) 20 (v) 30 (vi) 3 × 7, 4 × 8, 6 × 2, 5 × 4, 3 × 5.

Question: Exercise 9, Q1 and Q2: (Q1) Write the repeated addition in "times": 9 + 9 + 9 (given: 3 times 9), 6 + 6 + 6 + 6, 4 + 4 + 4 + 4 + 4, 2 + 2 + 2 + 2 + 2, 3 + 3 + 3 + 3 + 3 + 3. (Q2) Write the "times" in multiplication form: 7 times 2 (given: 7 × 2), 9 times 5, 4 times 6, 8 times 3, 5 times 4.

Answer: Step 1: Count how many times the repeated number has come. Step 2: Q1: 6 four times = 4 times 6; 4 five times = 5 times 4. Step 3: 2 five times = 5 times 2; 3 six times = 6 times 3. Step 4: Q2: 9 times 5 = 9 × 5; 4 times 6 = 4 × 6; 8 times 3 = 8 × 3; 5 times 4 = 5 × 4. Final answer: Q1: 3 times 9 (given), 4 times 6, 5 times 4, 5 times 2, 6 times 3; Q2: 7 × 2

(given), 9 × 5, 4 × 6, 8 × 3, 5 × 4.

Question: Exercise 10, Q1 and Q2: (Q1) Match the multiplication pairs in any order: 5 × 6 (pair 6 × 5 given), 3 × 7, 6 × 4, 8 × 5, 9 × 2. (Q2) Complete: 3 × 8 = ___ × 3, 9 × 4 = ___ × 9, 7 × ___ = 6 × 7, ___ × 6 = 6 × 2, 5 × 8 = 8 × ___.

Answer: Step 1: In multiplication changing the order of numbers does not change the answer, e.g. 5 rows of 4 balls = 20 and 4 columns of 5 = 20, so 5 × 4 = 4 × 5. Step 2: Q1: The pair of 3 × 7 is 7 × 3; of 6 × 4 is 4 × 6; of 8 × 5 is 5 × 8; of 9 × 2 is 2 × 9. Step 3: Q2: The missing number is the one absent from the other side: 3 × 8 = 8 × 3 (8); 9 × 4 = 4 × 9 (4). Step 4: 7 × 6 = 6 × 7 (6); 2 × 6 = 6 × 2 (2); 5 × 8 = 8 × 5 (5). Final answer: Q1: 5 × 6—6 × 5, 3 × 7—7 × 3, 6 × 4—4 × 6, 8 × 5—5 × 8, 9 × 2—2 × 9; Q2: 8, 4, 6, 2, 5.

Question: Division = repeated subtraction / equal sharing: (i) 8 pieces of cake, 2 in each plate — how many plates? (ii) 12 apples, 3 in each plate — how many plates? Show by repeated subtraction. (iii) Share 8 eggs equally in 4 baskets — how many in each basket? (iv) Can 4 eggs be divided equally in 8 baskets? Can division be done in any order?

Answer: Step 1: Division means equal sharing; its sign is "÷". Step 2: (i) Keep putting 2 pieces in each plate: taking 2 at a time from 8 fills 4 plates — 8 ÷ 2 = 4 plates. Step 3: (ii) Repeated subtraction: 12 − 3 = 9, 9 − 3 = 6, 6 − 3 = 3, 3 − 3 = 0 — 3 was subtracted 4 times in all, so 12 ÷ 3 = 4 plates. Step 4: (iii) 8 ÷ 4 = 2 — 2 eggs in each basket. Step 5: (iv) No, 4 eggs cannot come equally in 8 baskets; 8 ÷ 4 and 4 ÷ 8 are not the same — division cannot be done in any order. Final answer: (i) 4 plates (ii) 4 plates (iii) 2 eggs (iv) No; division does not change order.

Question: Exercise 11, Q1 and Q2: (Q1) 15 toffees, 3 in each basket — by repeated subtraction tell how many baskets are needed? (Q2) Tell

True/False: (a) 6 ÷ 2 is the same as 2 ÷ 6 (b) 8 ÷ 2 is not the same as 2 ÷ 8 (c) 10 ÷ 2 = 2 ÷ 10 (d) 18 ÷ 6 is not the same as 6 ÷ 18 (e) 15 ÷ 3 = 3 ÷ 15.

Answer: Step 1: Q1: Keep subtracting 3 from 15: 15 − 3 = 12 (given), 12 − 3 = 9, 9 − 3 = 6, 6 − 3 = 3, 3 − 3 = 0. Step 2: 3 was subtracted 5 times in all, so 15 ÷ 3 = 5 — 5 baskets are needed. Step 3: Q2 rule: In division changing the order changes the answer, so reversed pairs are never equal. Step 4: (a) ✗ False; (b) ✓ True — given; (c) ✗ False; (d) ✓ True; (e) ✗ False. Final answer: Q1: 5 baskets; Q2: (a) ✗ (b) ✓ (c) ✗ (d) ✓ (e) ✗.

Question: Exercise 12: Multiplication and division word problems: (1) One car has 4 wheels; total wheels of 5 cars? (given solved) (2) One bicycle has 2 wheels; total wheels of 6 bicycles? (3) One packet has 10 pencils; total pencils in 4 packets? (4) There are 10 books, each student gets 5 books — how many students can take the books? (5) There are 12 mangoes, 4 in each plate — how many plates are needed?

Answer: Step 1: (1) 4 + 4 + 4 + 4 + 4 = 5 times 4 = 20 wheels (given). Step 2: (2) 2 + 2 + 2 + 2 + 2 + 2 = 6 times 2 = 12 wheels. Step 3: (3) 10 + 10 + 10 + 10 = 4 times 10 = 40 pencils. Step 4: (4) Repeated subtraction: 10 − 5 = 5, 5 − 5 = 0 — 5 was subtracted 2 times, so 10 ÷ 5 = 2 students. Step 5: (5) 12 − 4 = 8, 8 − 4 = 4, 4 − 4 = 0 — 4 was subtracted 3 times, so 12 ÷ 4 = 3 plates. Final answer: (1) 20 wheels (2) 12 wheels (3) 40 pencils (4) 2 students (5) 3 plates.

Money

Question: Recognise coins and notes: Which are the Pakistani coins and which are the notes?

Answer: Step 1: Coins are made of metal, small and round; notes are made of paper and bigger. Step 2: Coins: 1-rupee coin = Rs. 1; 2-rupee coin = Rs. 2; 5-rupee coin = Rs. 5; 10-rupee coin = Rs. 10. Step 3: Notes: Rs. 10, Rs. 20, Rs. 50, Rs. 100 and Rs. 500 notes. Step 4: The value is written on every coin and note — read it to recognise. Final answer: Coins Rs. 1, Rs. 2, Rs. 5, Rs. 10; notes Rs. 10, Rs. 20, Rs. 50, Rs. 100, Rs. 500.

Question: Exercise 1, Q1 and Q2: (Q1) Match the coins with their values: Rs. 2 coin (given), Rs. 5, Rs. 10, Rs. 1 coins. (Q2) Match the notes with the values: Rs. 50, Rs. 500, Rs. 10, Rs. 20 (given), Rs. 100 notes.

Answer: Step 1: Read the number written on each coin/note and match it with the same value. Step 2: Q1: Rs. 5 coin — Rs. 5; Rs. 10 coin — Rs. 10; Rs. 1 coin — Rs. 1. Step 3: Q2: Rs. 50 note — Rs. 50; Rs. 500 note — Rs. 500. Step 4: Rs. 10 note — Rs. 10; Rs. 100 note — Rs. 100. Final answer: Q1: Rs. 2—Rs. 2 (given), Rs. 5—Rs. 5, Rs. 10—Rs. 10, Rs. 1—Rs. 1; Q2: Rs. 50—Rs. 50, Rs. 500—Rs. 500, Rs. 10—Rs. 10, Rs. 20—Rs. 20 (given), Rs. 100—Rs. 100.

Question: Activity: Tell the total price of two things: (i) Pencil Rs. 10 + candy Rs. 5 (given) (ii) Ice-cream Rs. 25 + ice lolly Rs. 15 (iii) Spinning top Rs. 20 + tennis ball Rs. 50 (iv) Chocolate Rs. 15 + toffee Rs. 5 (v) Biscuits Rs. 30 + cake Rs. 20.

Answer: Step 1: For the total price add the prices of both things — first the ones, then the tens. Step 2: (i) 10 + 5 = 15 rupees (given). Step 3: (ii) 25 + 15: ones 5 + 5 = 10 (write 0, carry 1 ten), tens 2 + 1 + 1 = 4 — 40

rupees. Step 4: (iii) 20 + 50 = 70 rupees; (iv) 15 + 5 = 20 rupees. Step 5: (v) 30 + 20 = 50 rupees. Final answer: (i) Rs. 15 (ii) Rs. 40 (iii) Rs. 70 (iv) Rs. 20 (v) Rs. 50.

Question: Activity: You have Rs. 99. After buying each thing, how much money will be left? (i) Party hat Rs. 50 (given) (ii) Ball Rs. 75 (iii) Glass Rs. 65 (iv) Pencil Rs. 10 (v) Fountain pen Rs. 90.

Answer: Step 1: Money left = your money − price of the thing. Step 2: (i) 99 − 50: ones 9 − 0 = 9, tens 9 − 5 = 4 — 49 rupees left (given). Step 3: (ii) 99 − 75: 9 − 5 = 4, 9 − 7 = 2 — 24 rupees. Step 4: (iii) 99 − 65 = 34 rupees; (iv) 99 − 10 = 89 rupees. Step 5: (v) 99 − 90 = 9 rupees. Final answer: (i) Rs. 49 (ii) Rs. 24 (iii) Rs. 34 (iv) Rs. 89 (v) Rs. 9.

Question: Exercise 2, Q1: Tell the total cost: (i) Eraser Rs. 5 + sharpener Rs. 10 (ii) Notebook Rs. 50 + pen Rs. 20 (iii) Balloon Rs. 15 + party hat Rs. 50 (iv) Popcorn Rs. 15 + ice lolly Rs. 20 (v) Carrot Rs. 10 + radish Rs. 20.

Answer: Step 1: Add both prices — ones then tens. Step 2: (i) 5 + 10 = 15 rupees. Step 3: (ii) 50 + 20 = 70 rupees. Step 4: (iii) 15 + 50 = 65 rupees; (iv) 15 + 20 = 35 rupees. Step 5: (v) 10 + 20 = 30 rupees. Final answer: (i) Rs. 15 (ii) Rs. 70 (iii) Rs. 65 (iv) Rs. 35 (v) Rs. 30.

Question: Exercise 2, Q2: Aafia has Rs. 95. After buying each thing, how much money will she have left? (i) Shuttlecock Rs. 65 (given) (ii) Comb Rs. 30 (iii) Glass Rs. 75 (iv) Mug Rs. 55 (v) Ruler Rs. 20.

Answer: Step 1: Money left = 95 − price of the thing. Step 2: (i) 95 − 65: ones 5 − 5 = 0, tens 9 − 6 = 3 — 30 rupees (given). Step 3: (ii) 95 − 30 = 65 rupees; (iii) 95 − 75: 5 − 5 = 0, 9 − 7 = 2 — 20 rupees. Step 4: (iv) 95 − 55 = 40 rupees; (v) 95 − 20 = 75 rupees. Final answer: (i) Rs. 30 (ii) Rs. 65 (iii) Rs. 20 (iv) Rs. 40 (v) Rs. 75.

Fractions

Question: Concept of fractions: (i) When a whole thing is divided into 2 equal parts, what is each part called? (ii) How do we write one part, two parts and three parts out of 4 equal parts? (iii) Colour one half of a circle, rectangle and triangle; colour a quarter of a square and rectangle; colour two quarters and three quarters of a square, rectangle and flower and write the fraction.

Answer: Step 1: A fraction is an equal part of a whole. Step 2: (i) Each part of 2 equal parts is a "half", written 1/2. Step 3: (ii) One part of 4 equal parts is a "quarter" = 1/4; two parts = two quarters = 2/4; three parts = three quarters = 3/4. Step 4: (iii) For a half, colour 1 of the 2 equal parts of the shape and write 1/2; for a quarter colour 1 of 4 parts and write 1/4. Step 5: For two quarters colour 2 of 4 parts and write 2/4, and for three quarters colour 3 parts and write 3/4. Final answer: (i) Half = 1/2 (ii) 1/4, 2/4, 3/4 (iii) Coloured parts: half 1/2, quarter 1/4, two quarters 2/4, three quarters 3/4.

Question: Exercise 1, Q1: Tick ✓ the whole objects and cross ✗ the fractions (parts): whole pizza, cake slice, whole pomegranate, half-cut watermelon, whole apple, orange slices, whole watermelon, part of a bread loaf.

Answer: Step 1: A thing that is complete and uncut is "whole"; a cut part is a fraction. Step 2: Whole pizza ✓; cake slice ✗ (it is a part). Step 3: Whole pomegranate ✓; half watermelon ✗ (half part). Step 4: Whole apple ✓; orange slices ✗; whole watermelon ✓. Step 5: Part of a bread loaf ✗ (it is shown as a fraction/part in the book). Final answer: ✓ whole pizza, whole pomegranate, whole apple, whole watermelon; ✗ cake slice, half

watermelon, orange slices, part of bread loaf.

Question: Exercise 1, Q2: Match the fractions with their names: Half, Quarter, Two quarters (given), Three quarters — fractions: 1/2, 1/4, 2/4, 3/4.

Answer: Step 1: Read the name and think: how many parts in all and how many parts taken? Step 2: Half means 1 part out of 2 = 1/2. Step 3: Quarter = 1 out of 4 = 1/4. Step 4: Two quarters = 2/4 (given); three quarters = 3/4. Final answer: Half—1/2, Quarter—1/4, Two quarters—2/4, Three quarters—3/4.

Question: Exercise 1, Q3 and Q4: (Q3) Colour one half and write the fraction: hexagon (1/2 given), a rectangle of 2 parts, 8 balls. (Q4) Colour the quarters: in the triangle shape 1/4, 2/4, 3/4 (given); in the 4-square shape 1, 2, 3 squares; out of 8 balls 2, 4, 6 balls — write the fraction of each.

Answer: Step 1: Q3: For a half, colour half of the total parts. Step 2: The rectangle has 2 parts — colour 1 = 1/2; half of 8 balls = colour 4 balls = 1/2. Step 3: Q4: Colouring 1 of 4 squares = 1/4; 2 = 2/4; 3 = 3/4. Step 4: Colouring 2 of 8 balls makes one quarter of the 8 balls = 1/4; 4 balls = 2/4; 6 balls = 3/4. Final answer: Q3: 1/2, 1/2, 1/2 (4 balls); Q4: 1/4, 2/4, 3/4; for balls 1/4 (2 balls), 2/4 (4 balls), 3/4 (6 balls).

Question: Exercise 1, Q5 and Q6: (Q5) Match the fractions with the correct picture: 1/4 (given), 2/4, 1/2, 3/4. (Q6) Write the fraction of the coloured part of each picture: triangle (1 of 4 parts green), hexagon (3 of 4 blue), circles shape (2 of 4), rectangle (1 of 2 green), big triangle (3 of 4 orange), triangles strip (2 of 4 red), kite (1 of 4 blue), kite (1 of 2 orange), kite (3 of 4 blue), kite (2 of 4 red).

Answer: Step 1: In each picture first count the total equal parts (bottom number), then the coloured parts (top number). Step 2: Q5: 2/4 — the

picture with 2 of 4 squares coloured; 1/2 — 1 of 2 parts coloured; 3/4 — 3 of 4 coloured. Step 3: Q6: Triangle = 1/4; hexagon = 3/4; circles = 2/4; rectangle = 1/2. Step 4: Big triangle = 3/4; strip = 2/4; kites = 1/4, 1/2, 3/4, 2/4. Final answer: Q5: 1/4 (given), 2/4, 1/2, 3/4; Q6: 1/4, 3/4, 2/4, 1/2, 3/4, 2/4, 1/4, 1/2, 3/4, 2/4.

Question: Numerator and denominator, and adding/subtracting fractions with the same denominator: (i) Tell the numerator and denominator in 2/4, 3/6 and 5/8. (ii) Add: 1/4 + 2/4, 2/7 + 3/7, 4/8 + 2/8. (iii) Subtract: 3/5 − 1/5, 4/7 − 3/7, 5/8 − 2/8.

Answer: Step 1: In a fraction the top number = parts taken (numerator); the bottom number = total equal parts (denominator). Step 2: (i) 2/4: numerator 2, denominator 4; 3/6: numerator 3, denominator 6; 5/8: numerator 5, denominator 8. Step 3: (ii) Rule: when the denominator is the same, add only the numerators: 1/4 + 2/4 = 3/4; 2/7 + 3/7 = 5/7; 4/8 + 2/8 = 6/8. Step 4: (iii) In subtraction also subtract only the numerators: 3/5 − 1/5 = 2/5; 4/7 − 3/7 = 1/7; 5/8 − 2/8 = 3/8. Final answer: (i) 2/4, 3/6, 5/8 (ii) 3/4, 5/7, 6/8 (iii) 2/5, 1/7, 3/8.

Question: Exercise 2, Q1 and Q2: (Q1) Add with pictures: 3/5 + 1/5 (4/5 given), 2/4 + 1/4, 2/5 + 1/5, 3/6 + 2/6. (Q2) Match the additions: 2/5 + 1/5 (3/5 given), 3/7 + 2/7, 1/4 + 2/4, 2/6 + 3/6.

Answer: Step 1: The denominators are the same — add only the numerators and write the same denominator. Step 2: Q1: 2/4 + 1/4: 2 + 1 = 3 — 3/4; 2/5 + 1/5 = 3/5; 3/6 + 2/6 = 5/6. Step 3: Q2: 3/7 + 2/7 = 5/7; 1/4 + 2/4 = 3/4; 2/6 + 3/6 = 5/6. Final answer: Q1: 4/5 (given), 3/4, 3/5, 5/6; Q2: 3/5 (given), 5/7, 3/4, 5/6.

Question: Exercise 2, Q3 and Q4: (Q3) Subtract with pictures: 3/5 − 1/5 (2/5 given), 4/6 − 2/6, 4/6 − 1/6. (Q4) Match the subtractions: 5/7 − 2/7 (3/7 given), 4/5 − 1/5, 6/8 − 2/8, 5/9 − 3/9.

Answer: Step 1: The denominators are the same — subtract only the numerators. Step 2: Q3: 4/6 − 2/6: 4 − 2 = 2 — 2/6; 4/6 − 1/6 = 3/6. Step 3: Q4: 4/5 − 1/5 = 3/5; 6/8 − 2/8 = 4/8; 5/9 − 3/9 = 2/9. Final answer: Q3: 2/5 (given), 2/6, 3/6; Q4: 3/7 (given), 3/5, 4/8, 2/9.

Algebra

Question: Concept of pattern, examples and activity: A pattern is a special arrangement of things in a repeated order. In each row a pair repeats — tell what comes next: (i) Example: the pair (football, tennis ball) is repeated three times, the row ends on tennis ball — what is next? (✓ given in book) (ii) Example: the pair (chair, table) three times and then chair — what is next? (given) (iii) Activity: the pair (triangle-face, round-face) two times and then triangle-face — what is next? (iv) The pair (apple, banana) three times — what is next? (v) The pair (eraser, pencil) four times and then eraser — what is next?

Answer: Step 1: In a pattern a pair of two things repeats again and again: A, B, A, B... The next thing is the one that completes the pair. Step 2: (i) The row ends on tennis ball (B), a new pair begins — next is football (A) ✓ given. Step 3: (ii) The row ends on chair (A), to complete the pair next is table (B) ✓ given. Step 4: (iii) After the triangle-face comes the round-face. Step 5: (iv) The pair (apple, banana) is complete, a new pair starts with apple — next is apple; (v) after eraser, to complete the pair, comes pencil. Final answer: (i) football (ii) table (iii) round-face (iv) apple (v) pencil.

Question: Exercise 1: In each row a pair repeats — tell the next object: (i) (round-face, cube) (ii) (football, bat) (iii) (triangle, cube) (iv) (mango, pomegranate) (v) (pencil, eraser) (vi) (mug, glass).

Answer: Step 1: The rule of the book's rows is: the object that has not come at the end of the row comes next — the pair keeps alternating. Step 2: (i) The row ends on cube, next is round-face; (ii) the row ends on football, next is bat. Step 3: (iii) The row ends on triangle, next is cube; (iv) the row ends on mango, next is pomegranate. Step 4: (v) The row ends on

pencil, next is eraser; (vi) the row ends on glass, next is mug. Final answer: round-face, bat, cube, pomegranate, eraser, mug.

Question: Exercise 2 and the colour/size activity: (Activity) (i) Pair (yellow ball, green ball) — next? (ii) (tall bamboo stack, short stack) — next? (iii) (big spoon, small spoon) — next? (Exercise 2) (iv) (red ball, blue ball) — next? (v) (red rose, yellow rose) — next? (vi) (big bat, small bat) — next? (vii) Letter patterns: P Q P Q P Q P Q — next?; L M L M L M L M L — next?; D F D F D F D — next?

Answer: Step 1: The same pair rule: after the last object of the row comes the other object of the pair — check the end of each row in the book. Step 2: Answers as in the book: (i) green ball (ii) short stack (iii) big spoon. Step 3: (iv) red ball (v) yellow rose (vi) small bat. Step 4: (vii) Letters repeat the same way: in the P Q repeat the row ends on Q — next is P; in L M the row ends on L — next is M; in D F the row ends on D — next is F. Final answer: (i) green ball (ii) short stack (iii) big spoon (iv) red ball (v) yellow rose (vi) small bat (vii) P; M; F.

Measurement

Question: Comparing length and height: (i) Who is taller and who is shorter, Sara or Ali? (ii) Which pencil is longer, the red or the blue? (iii) Name the three rulers (6, 8, 10) with Long, Longer, Longest and tall things with Tall, Taller, Tallest.

Answer: Step 1: Stand/place the two things side by side and see which is longer/taller. Step 2: (i) Sara is taller than Ali; Ali is shorter than Sara. Step 3: (ii) The red pencil is longer than the blue pencil; the blue one is shorter. Step 4: (iii) In order from smallest to biggest: first Long, second Longer, third and longest Longest; for height: Tall, Taller, Tallest (for short: Short, Shorter, Shortest). Final answer: (i) Sara taller, Ali shorter (ii) red longer (iii) Long, Longer, Longest; Tall, Taller, Tallest.

Question: Measuring with non-standard units: (i) How many paper clips long is the book? The pencil? Which thing is longer? (ii) Tell the lengths of the desk and the blackboard in crayons — which is shorter? (iii) Tell the weight of the marker and the ball in marbles — which is lighter? (iv) Tell the capacity of the bottle and the jug in glasses — which has more capacity?

Answer: Step 1: In non-standard measuring, a small thing is made the unit and counted along the big thing. Step 2: (i) Book = 5 paper clips long; pencil = 4 clips; the book is longer than the pencil (5 > 4). Step 3: (ii) Desk = 5 crayons; blackboard = 7 crayons; the desk is shorter than the board (5 < 7). Step 4: (iii) Marker = 2 marbles; ball = 4 marbles; the marker is lighter than the ball and the ball is heavier. Step 5: (iv) Bottle = 4 glasses; jug = 5 glasses; the jug has more capacity than the bottle. Final answer: (i) book 5, pencil 4 — book is longer (ii) desk 5, board 7 — desk is shorter

(iii) marker 2, ball 4 — marker is lighter (iv) bottle 4, jug 5 — jug has more.

Question: Exercise 1: (Q1) Tick ✓ the taller animal/child and cross ✗ the shorter: big camel and small camel (given), big and small zebra, taller and shorter girl. (Q2) Match the buildings with Tall, Taller, Tallest. (Q3) Match the flutes with Long, Longer, Longest.

Answer: Step 1: Between two things of the same kind, tick ✓ the taller/longer one and cross ✗ the shorter one. Step 2: Q1: ✓ on the big camel and ✗ on the small one (given); ✓ big zebra, ✗ small zebra; ✓ taller girl, ✗ shorter girl. Step 3: Q2: Of the three buildings the tallest = Tallest, the middle = Taller, the shortest = Tall. Step 4: Q3: The longest flute = Longest, the middle = Longer, the shortest = Long. Final answer: Q1: ✓ on big camel, big zebra, taller girl; ✗ on the smaller ones. Q2: Tallest, Taller, Tall. Q3: Longest, Longer, Long.

Question: Exercise 2: (Q1) Tick ✓ the heavier thing and cross ✗ the lighter: rose and book (given), jug of water and mug, cricket bat and tennis ball, rabbit and cat. (Q2) Match mobile phone, laptop and desktop computer with Heavy, Heavier, Heaviest. (Q3) Match tennis ball, balloon and kite with Light, Lighter, Lightest. (Note: a watermelon is heavier than an apple and an apple is lighter than a watermelon; horse = Heavy, camel = Heavier, elephant = Heaviest.)

Answer: Step 1: In each pair the heavier thing is the one that pulls the balance down more or feels heavier to lift. Step 2: Q1: Book is heavier than rose ✓ (given), rose ✗; jug of water ✓, mug ✗; cricket bat ✓, tennis ball ✗; rabbit ✓, cat ✗. Step 3: Q2: Mobile phone = Heavy, laptop = Heavier, desktop computer = Heaviest (a bigger thing is usually heavier). Step 4: Q3: Tennis ball = Light, balloon = Lighter, kite = Lightest. Final answer: Q1: ✓ book, jug, bat, rabbit; ✗ rose, mug, tennis ball, cat. Q2: Heavy, Heavier, Heaviest. Q3: Light, Lighter, Lightest.

Question: Exercise 3: (Q1) Tick ✓ the thing with more capacity and cross ✗ the one with less: mug and kettle (given), bottle and glass, bucket and bowl. (Q2) Match the three bottles (small, medium, big) with Large, Larger, Largest. (Note: glass = Large, jug = Larger, bucket = Largest; a bucket holds more water than a glass.)

Answer: Step 1: Capacity means how much water/material a thing can hold — a bigger thing has more capacity. Step 2: Q1: Kettle holds more than mug ✓ (given), mug ✗; bottle ✓, glass ✗; bucket ✓, bowl ✗. Step 3: Q2: Small bottle = Large, medium = Larger, biggest = Largest. Final answer: Q1: ✓ kettle, bottle, bucket; ✗ mug, glass, bowl. Q2: Large, Larger, Largest.

Question: Exercise 4: Seasons: (Q1) Tick ✓ the correct season and cross ✗ the wrong one: Spring and Summer (✓ on Summer given); Autumn and Winter; Spring and Winter; Summer and Autumn — in each pair recognise the season in the picture. (Q2) Match the season with its warmth: Spring, Winter, Summer, Autumn — Warm, Coolest, Warmest, Cool (given).

Answer: Step 1: Recall the signs of the seasons: in Spring the weather gets warmer and new leaves and flowers come out; Summer is the warmest and sunniest; in Autumn it gets cooler and leaves fall; Winter is the coldest. Step 2: Q1: In the given pair Summer ✓, Spring ✗ (the picture is of summer). Step 3: Other pairs: Autumn ✓, Winter ✗; Spring ✓, Winter ✗; Summer ✓, Autumn ✗ — each time ✓ on the pictured season. Step 4: Q2: Spring = Warm, Winter = Coolest, Summer = Warmest, Autumn = Cool (given). Final answer: Q1: ✓ on the Summer, Autumn, Spring, Summer pictures; Q2: Spring—Warm, Winter—Coolest, Summer—Warmest, Autumn—Cool.

Question: Time: (i) In an analog clock what do the short hand and the long hand show? If the minute hand is at 12 and the hour hand is at 9, 8, 2, 5,

3, 11, what is the time? (ii) Tell the meaning of the digital clock: 05:00, 06:00, 10:00, 03:00, 02:00, 07:00, 04:00. (iii) Activity: Show 4 o'clock (given), 10 o'clock, 7 o'clock, 5 o'clock on analog clocks; write 6 o'clock (06:00 given), 9 o'clock, 11 o'clock, 8 o'clock in digital.

Answer: Step 1: (i) The short hand is the hour hand; the long hand is the minute hand. Step 2: When the minute hand is at 12, it is a full hour — read the number at the hour hand: 9 o'clock, 8 o'clock, 2 o'clock, 5 o'clock, 3 o'clock, 11 o'clock. Step 3: (ii) In digital, the first part is hours and the second is minutes: 05:00 = 5 o'clock; 06:00 = 6 o'clock; 10:00 = 10 o'clock; 03:00 = 3 o'clock; 02:00 = 2 o'clock; 07:00 = 7 o'clock; 04:00 = 4 o'clock. Step 4: (iii) To show a time, put the hour hand on the needed number and the minute hand on 12: for 4 o'clock the hand is at 4 (given), at 10, at 7, at 5. Step 5: Digital: 9 o'clock = 09:00; 11 o'clock = 11:00; 8 o'clock = 08:00. Final answer: (i) short = hours, long = minutes; 9, 8, 2, 5, 3, 11 o'clock (ii) 5, 6, 10, 3, 2, 7, 4 o'clock (iii) hands at 4, 10, 7, 5; 06:00 (given), 09:00, 11:00, 08:00.

Question: Exercise 5: (Q1) What time do the analog clocks show (three clocks) and read the digital: 05:00, 10:00, 09:00. (Q2) Show 2, 5, 1 o'clock on analog clocks; write 6, 4, 11 o'clock in digital. (Q3) Match the analog clocks with digital: the 5 o'clock one, 8 o'clock, 10 o'clock, 4 o'clock.

Answer: Step 1: Q1: Read the number at the hour hand (the minute hand is at 12): first clock 2 o'clock, second 7 o'clock, third 4 o'clock. Step 2: Digital: 05:00 = 5 o'clock; 10:00 = 10 o'clock; 09:00 = 9 o'clock. Step 3: Q2: To show, put the short hand on the number and the long hand on 12: 2 o'clock, 5 o'clock, 1 o'clock; digital: 6 = 06:00, 4 = 04:00, 11 = 11:00. Step 4: Q3: 5 o'clock — 05:00; 8 o'clock — 08:00; 10 o'clock — 10:00; 4 o'clock — 04:00. Final answer: Q1: 2, 7, 4 o'clock; 5, 10, 9 o'clock. Q2: hands at 2, 5, 1; 06:00, 04:00, 11:00. Q3: 5—05:00, 8—08:00, 10—10:00,

4—04:00.

Question: Days and months: (i) Tell the ordinal positions of the 7 days of the week. (ii) Tell the day before Thursday (given), before Friday, before Sunday; the day after Monday (given), after Friday, after Saturday. (iii) Tell the names and order of the 12 months of the year; the month before April (given), before December, before May; the month after January (given), after August, after October.

Answer: Step 1: (i) Days: Monday 1st, Tuesday 2nd, Wednesday 3rd, Thursday 4th, Friday 5th, Saturday 6th, Sunday 7th. Step 2: (ii) Before = one step back: before Thursday is Wednesday (given); before Friday is Thursday; before Sunday is Saturday. Step 3: After = one step forward: after Monday is Tuesday (given); after Friday is Saturday; after Saturday is Sunday. Step 4: (iii) Months: January 1st, February 2nd, March 3rd, April 4th, May 5th, June 6th, July 7th, August 8th, September 9th, October 10th, November 11th, December 12th. Step 5: Months before: before April is March (given); before December is November; before May is April. After: after January is February (given); after August is September; after October is November. Final answer: (i) 1st–7th as above (ii) Wednesday, Thursday, Saturday; Tuesday, Saturday, Sunday (iii) March, November, April; February, September, November.

Question: Exercise 6: (Q1) Match the days with their ordinal positions: Monday, Tuesday, Wednesday, Thursday, Friday (5th given), Saturday, Sunday. (Q2) Fill in the missing days: Monday, ___, Wednesday; ___, Sunday, Monday; Wednesday, Thursday, ___; ___, Friday, ___. (Q3) Match the months with their positions: January, February, March, July, August, October (10th given).

Answer: Step 1: The days of the week in order: Monday, Tuesday, Wednesday, Thursday, Friday, Saturday, Sunday. Step 2: (Q1) Monday 1st,

Tuesday 2nd, Wednesday 3rd, Thursday 4th, Friday 5th (given), Saturday 6th, Sunday 7th. Step 3: (Q2) After Monday comes Tuesday; before Sunday is Saturday; after Thursday comes Friday; before Friday is Thursday and after it is Saturday. Step 4: (Q3) January is the 1st month, February 2nd, March 3rd, July 7th, August 8th and October the 10th (given). Final answer: (Q1) 1st, 2nd, 3rd, 4th, 5th, 6th, 7th. (Q2) Tuesday; Saturday; Friday; Thursday, Saturday. (Q3) 1st, 2nd, 3rd, 7th, 8th, 10th.

Geometry

Question: Exercise 1, Q1: Tell the sides and corners of each 2-D shape: Rectangle (given), Square, Circle, Triangle.

Answer: Step 1: Sides are the straight/round boundary lines of the shape; corners are where two sides meet. Step 2: Rectangle: 4 sides, 4 corners (given) — opposite sides are equal (e.g. blackboard). Step 3: Square: 4 sides, 4 corners — all sides are equal (e.g. white board). Step 4: Circle: 0 sides, 0 corners — a perfectly round shape (e.g. CD). Step 5: Triangle: 3 sides, 3 corners (e.g. road sign). Final answer: Rectangle 4/4, Square 4/4, Circle 0/0, Triangle 3/3.

Question: Exercise 1, Q2, Q3 and Q4: (Q2) What shape are these real-life things: a 100-rupee note, a wall clock, a road sign (children crossing), a carrom board. (Q3) Match: chocolate bar, triangular tile, CD, white board. (Q4) Colour the six pictures: colour every rectangle RED, every square BLUE, every circle GREEN and every triangle YELLOW.

Answer: Step 1: Compare the shape of the thing with the 2-D shapes. Step 2: Q2: The 100 note is long and not a square — Rectangle; the wall clock is round — Circle. Step 3: The road sign has three sides — Triangle; the carrom board has all sides equal — Square. Step 4: Q3: Chocolate bar — Rectangle; triangular tile — Triangle; CD — Circle; white board — Square. Step 5: Q4: Recall the rule and colour: rectangle = red, square = blue, circle = green, triangle = yellow. Final answer: Q2: Rectangle, Circle, Triangle, Square; Q3: Rectangle, Triangle, Circle, Square; Q4: red/blue/green/yellow in that order.

Question: Exercise 2, Q1 and Q2: 3-D shapes: (Q1) What 3-D shape are these real things: dice, matchbox, football, battery cell, party hat. (Q2)

Match: stacking-rings toy (given), basketball, puzzle cube, sharpener, drink can. (Note: A cube has 6 faces and 8 corners; a cuboid also has 6 faces, 8 corners; a cone has 1 circular face and 1 corner; a sphere has no faces and no corners; a cylinder has 2 circular faces and no corners.)

Answer: Step 1: Touch/look at the thing and recognise which 3-D shape it is like. Step 2: Q1: A dice is an equal box from all sides — Cube; a matchbox is a long box — Cuboid. Step 3: A football is perfectly round — Sphere; a battery cell — Cylinder; a party hat is pointed — Cone. Step 4: Q2: Stacking-rings toy — Cone (given); basketball — Sphere; puzzle cube — Cube; sharpener — Cuboid; can — Cylinder. Final answer: Q1: Cube, Cuboid, Sphere, Cylinder, Cone; Q2: Cone (given), Sphere, Cube, Cuboid, Cylinder.

Question: Exercise 3, Q1: "Inside/outside": Tick ✓ the correct and cross ✗ the wrong: dog inside the kennel (✓ given), dog outside the kennel (✗ given), ball outside the hoop, ball inside the net, coins inside the jar, coins outside the jar, bird outside the cage, bird inside the cage.

Answer: Step 1: In each picture see whether the thing is inside or outside the closed place. Step 2: Dog inside the kennel ✓ (given); dog outside ✗ (given — because the sentence was for "inside" and the dog is outside). Step 3: Ball outside the hoop ✗; ball inside the net ✓. Step 4: Coins inside the jar ✓; coins outside the jar ✗; bird outside the cage ✗; bird inside the cage ✓. Final answer: ✓, ✗, ✗, ✓, ✓, ✗, ✗, ✓.

Question: Exercise 3, Q2 and Q3: (Q2) Parrot "above/below": above the tree (✓ given), below the clouds (✗ given)... the rest: below the aeroplane, above the house, above the car, below the hot-air balloon. (Q3) Dog "over/under": over the tree stump (✓ given), under the car (✗ given), under the tree, over the ball, over the box, under the table.

Answer: Step 1: "Above/over" means on the higher side of a thing;

"below/under" means on the lower side. Step 2: In Q2 the book's pattern marks the "above" sentences ✓ and the "below" ones ✗; as per the given answers, in the parrot pictures: above the tree ✓, below the clouds ✗, below the aeroplane ✗, above the house ✓, above the car ✓, below the balloon ✗. Step 3: Q3: over the stump ✓ (given), under the car ✗ (given), under the tree ✗, over the ball ✓, over the box ✓, under the table ✗. Final answer: Q2: ✓, ✗, ✗, ✓, ✓, ✗; Q3: ✓, ✗, ✗, ✓, ✓, ✗.

Question: Exercise 3, Q4 and Q5: (Q4) Circle the correct thing: the near house (circled given), the far car (circled given), the far boat, the near camel, the near eagle, the far tree. (Q5) Puppy right/left: left of the apple (✗ given), right of the basket (✓ given), right of the handbag, left of the duck, left of the jug, left of the globe, right of the bowling pin, right of the teddy bear — ✓ for right, ✗ for left.

Answer: Step 1: Q4: A "near" thing looks big and in front; a "far" thing looks small and distant. Step 2: Given: the big house in front is near ✓ circled; the small car on the road is far ✓ circled. Step 3: The rest: the small boat is far — circle it; the big camel is near — circle; the big eagle is near — circle; the small palm tree on the hill is far — circle. Step 4: Q5: Treat the side of your right hand as "right": right of basket ✓ (given), right of handbag ✓, right of bowling pin ✓, right of teddy ✓; left of apple ✗ (given), left of duck ✗, left of jug ✗, left of globe ✗. Final answer: Q4: near house, far car (given), far boat, near camel, near eagle, far tree; Q5: ✗, ✓, ✓, ✗, ✗, ✗, ✓, ✓.

Statistics and Probability

Question: Pictograph — Favorite fruit: Read the graph and tell: how many students like mango, apple, banana, grapes and guava? Which fruit is the most favourite and which is the least favourite?

Answer: Step 1: A pictograph shows information with pictures — count the faces/pictures in front of each fruit. Step 2: Mango = 3, apple = 2. Step 3: Banana = 5, grapes = 4, guava = 1. Step 4: The biggest number is 5 — banana is the most favourite; the smallest is 1 — guava is the least favourite. Final answer: Mango 3, apple 2, banana 5, grapes 4, guava 1; most: banana; least: guava.

Question: Activity: Pictograph of "Mode of transport": Tell the number of students using school bus, car, bicycle and motorcycle. Which mode is used the most and which the least?

Answer: Step 1: Count the faces in each row one by one. Step 2: School bus = 9, car = 3. Step 3: Bicycle = 6, motorcycle = 5. Step 4: The biggest is 9 — the school bus is used the most ✓; the smallest is 3 — the car is used the least ✓. Final answer: Bus 9, car 3, bicycle 6, motorcycle 5; most: school bus; least: car.

Question: Exercise 1, Q1 and Q2: (Q1) Fruit graph: Tell the numbers for pomegranate, watermelon, peach, orange, apricot; tick ✓ the most favourite and cross ✗ the least favourite. (Q2) Animal graph: Tell the numbers for horse, lion, goat, bear, zebra; tick ✓ the most popular and cross ✗ the least popular.

Answer: Step 1: Q1: Count the pictures in each row: pomegranate = 6, watermelon = 4. Step 2: Peach = 7, orange = 8, apricot = 3. Step 3: The biggest number 8 — orange is the most favourite ✓; the smallest 3 —

apricot is the least ✗. Step 4: Q2: Horse = 8, lion = 9, goat = 5, bear = 3, zebra = 6. Step 5: Lion is the most popular ✓ (9); bear is the least ✗ (3). Final answer: Q1: 6, 4, 7, 8, 3; ✓ orange, ✗ apricot. Q2: 8, 9, 5, 3, 6; ✓ lion, ✗ bear.

Question: Exercise 2: Likelihood (more likely/less likely): (Example) Likelihood of rain: under dark clouds (✓ given) and under a clear sunny sky (✗ given). (Q1) Getting late: in a heavy traffic jam and on an empty clear road. (Q2) Getting sick: a child wet in rain eating an ice lolly and a healthy child playing in the park. (Q3) Getting weak eyesight: using a mobile in dim/dark light and a child reading in good light/playing/sleeping. For each, tick ✓ "more likely" and cross ✗ "less likely".

Answer: Step 1: An event with more chance of happening is "more likely"; with less chance is "less likely". Step 2: Example: With dark clouds, rain is more likely ✓; with a clear sky, less likely ✗. Step 3: Q1: In a traffic jam getting late is more likely ✓; on an empty road less likely ✗. Step 4: Q2: For the child wet in rain, getting sick is more likely ✓; for the healthy playing child less likely ✗. Step 5: Q3: Using a mobile in the dark makes weak eyesight more likely ✓; reading in good light/playing/sleeping makes it less likely ✗. Final answer: Q1: ✓ traffic jam, ✗ empty road; Q2: ✓ wet child, ✗ healthy child; Q3: ✓ mobile in dark, ✗ good-light habits.

Hands-On STEAM Activities

Question: Hands-On STEAM Activities (teacher-led): (i) In the Fruit Shop game, buying 2 apples and 3 apples makes how many apples in all? (ii) In the Bowling game 10 cups stand as pins; if some fall down, how do we find the remaining? (iii) In the "Time Hunt" game, how do we recognise the 3 o'clock clock?

Answer: Step 1: (i) Count the bought apples together: 2 + 3 — count 3 forward from 2: 3, 4, 5. Step 2: Answer: 2 + 3 = 5 apples. Step 3: (ii) Remaining pins = total 10 − fallen pins; for example if 4 fell, 10 − 4 = 6 remaining. Step 4: (iii) At 3 o'clock the short hour hand is at 3 and the long minute hand is at 12 — find such a clock. Final answer: (i) 5 apples (ii) remaining = 10 − fallen (iii) short hand at 3, long hand at 12.

AI-Based Puzzle/Game Activities

Question: AI-Based Activities (teacher-led): (i) Riddle: "You have 3 apples and AI gives you 2 more — how many now?" (ii) Digital Domino: one side has 4 dots and the other side 2 dots — total dots? (iii) AI Balance: "Is an apple heavier than a pencil?" (iv) Cash Register game: How will you make Rs. 20 and Rs. 50?

Answer: Step 1: (i) Add both groups: 3 + 2 = 5. Step 2: (ii) Add the dots of both sides: 4 + 2 = 6. Step 3: (iii) Lift the apple in your hand — an apple is heavier than a pencil, so the answer is: Yes, the apple is heavier. Step 4: (iv) To make Rs. 20: Rs. 10 + Rs. 10 (two ten-rupee notes/coins); for Rs. 50: one Rs. 50 note. Final answer: (i) 5 apples (ii) 6 dots (iii) Yes, an apple is heavier than a pencil (iv) 10 + 10 = 20; one 50 note = 50.

Some Mathematical Websites/Links

Question: Which websites are given in the book for grade 1 mathematics practice?

Answer: Step 1: Read the list given at the end of the book. Step 2: The websites are: Khan Academy, IXL, Super Teacher Worksheets, K5 Learning. Step 3: Besides these: Math Playground, SplashLearn, Math Games, ABCya. Step 4: Open the grade 1 maths pages on these sites and practise with the guidance of a teacher/parents. Final answer: Khan Academy, IXL, Super Teacher Worksheets, K5 Learning, Math Playground, SplashLearn, Math Games, ABCya.

Mathematics Workbook

Question: Workbook — Sample and Exercises 1 to 4: (Sample) After 1 + 4 (5 given): 2 + 4, 3 + 4, 4 + 4, 7 + 0. (Ex1) Trace the dotted lines — how many lines are there? (Ex2) Trace the dotted shapes: square, square, circle, oval, triangle, triangle — how many shapes? (Ex3) Colour circles equal to the goats: 1, 2, 3, 4, 5. (Ex4 and Assessment 1) Write the number of fingers on the hands: one to five.

Answer: Step 1: Sample: Count forward from the smaller number: 2 + 4 = 6; 3 + 4 = 7; 4 + 4 = 8; 7 + 0 = 7 (adding zero keeps the number the same). Step 2: Ex1: The dotted lines are 18 in all — run the pencil over them. Step 3: Ex2: The shapes are 6 in all — trace square, square, circle, oval, triangle, triangle. Step 4: Ex3: In each row colour as many circles as the goats: 1, 2, 3, 4, 5. Step 5: Ex4/Assessment 1: Count the fingers and write: 1, 2, 3, 4, 5. Final answer: Sample: 6, 7, 8, 7; Ex1: 18 lines; Ex2: 6 shapes; Ex3: 1–5 circles; Ex4: 1, 2, 3, 4, 5.

Question: Workbook — Exercises 5 to 12 and Assessments 2, 3: (Ex5 and Assessment 2) Write the fingers on the hands from six to ten; in Assessment 2 count the things in the box: chairs and shirts. (Ex6) Count and write: trousers, drinking glasses, black circles, black circles. (Ex7) Count and write: ants, chilies, pigeons, coconuts. (Ex8) Descending stairs: 9, 8, ___, 6, ___, 4, 3, ___, ___. (Ex9) What number does the empty bowl show? (Ex10) Number bonds of 5: 5 = ___ + ___ (all forms). (Ex11) Missing numbers: 1, 2, 3, ___, 5; ___, 8, ___, 6; 2, ___, ___, 5, 6. (Ex12 and Assessment 3) Tick the bigger number: apples 5 and candies 4 (example), chairs 2 and desk 1, pears 4 and tomatoes 3; in Assessment 3 the bonds of 5 and the bigger number: from 2 and 7, from 9 and 6.

Answer: Step 1: Ex5/Assessment 2: Count the fingers and write: 6, 7, 8, 9, 10; in the box chairs = 3 and shirts = 8 (as per the book's answer key). Step 2: Ex6: Trousers 3, glasses 4, black circles 2, black circles 3. Step 3: Ex7: Ants 3, chilies 5, pigeons 6, coconuts 4. Step 4: Ex8: Backward counting: 9, 8, 7, 6, 5, 4, 3, 2, 1 — missing 7, 5, 2, 1. Step 5: Ex9: The empty bowl has nothing = 0, read as "zero" — trace and copy 0. Step 6: Ex10: 5 = 3 + 2; 5 = 1 + 4; 5 = 2 + 3; 5 = 1 + 4; 5 = 3 + 2; 5 = 2 + 3. Step 7: Ex11: 4; in descending 9, 7; 3, 4. Step 8: Ex12: Tick the bigger: 5 (example), 2, 4. Assessment 3: 5 = 2 + 3; 5 = 0 + 5; bigger numbers: 7 (of 2 and 7), 9 (of 9 and 6). Final answer: 6–10; chairs 3, shirts 8; 3, 4, 2, 3; 3, 5, 6, 4; 7, 5, 2, 1; 0; bonds of 5 above; 4; 9, 7; 3, 4; ✓ 5, 2, 4; 7, 9.

Question: Workbook — Exercises 13 to 20 and Assessments 4, 5: (Ex13) 9 and 1 make how much? 10 ones make what? (Ex14) Bonds of 10: 10 = 6 + ___, 8 + ___, 1 + ___, 2 + ___. (Ex15) Bonds of 10: 4 + ___, 3 + ___, 8 + ___, 9 + ___. (Ex16) Block numbers: example 14; then four numbers. (Assessment 4) 10 = 8 + ___; 10 = 9 + ___; number line: 2, ___, ___, 5, 6. (Ex17) 10 + 2, 10 + 3, 10 + 0, 10 + 9. (Ex18) Split: 13, 12, 14, 10 into 10 + ___. (Ex19) Number lines: missing in 0–10; two missing in 10–20. (Ex20) Blocks: example 21; then two numbers. (Assessment 5) Three missing in the 0–20 line; 6 + 10; 10 + 10; the block number.

Answer: Step 1: Ex13: 9 + 1 = 10; 10 ones = 1 ten — trace and copy "10/ten". Step 2: Ex14: Missing partner = 10 − given: 4, 2, 9, 8. Step 3: Ex15: 6, 7, 2, 1. Step 4: Ex16: Count the tens blocks and ones: 11, 13, 19, 15 (as per answer key). Step 5: Assessment 4: 10 = 8 + 2; 10 = 9 + 1; line: 3, 4. Step 6: Ex17: 10 + 2 = 12; 10 + 3 = 13; 10 + 0 = 10; 10 + 9 = 19. Step 7: Ex18: 13 = 10 + 3; 12 = 10 + 2; 14 = 10 + 4; 10 = 10 + 0. Step 8: Ex19: Missing in 0–10 line is 4; in 10–20 are 11 and 16. Step 9: Ex20: Blocks: 22, 23. Assessment 5: Missing in line 2, 8, 14; 6 + 10 = 16;

10 + 10 = 20; blocks = 24 (as per key). Final answer: 10; 1 ten; 4, 2, 9, 8; 6, 7, 2, 1; 11, 13, 19, 15; 2; 1; 3, 4; 12, 13, 10, 19; 3, 2, 4, 0; 4; 11, 16; 22, 23; 2, 8, 14; 16; 20; 24.

Question: Workbook — Block numbers (Exercises 21 to 28 and Assessments 6, 7): In each exercise an example is given and two numbers are to be counted: Ex21 (example 31), Ex22 (example 41), Ex23 (example 51), Ex24 (example 61), Ex25 (example 71), Ex26 (example 81), Ex27 (example 91), Ex28 (example 100); tell the four numbers of Assessment 6 and the four of Assessment 7.

Answer: Step 1: In each picture first count the tens blocks (each = 10), then the ones blocks. Step 2: Ex21: 33, 36; Ex22: 47, 43. Step 3: Ex23: 56, 54; Ex24: 63, 66. Step 4: Assessment 6: 37, 47, 40, 62. Step 5: Ex25: 73, 77; Ex26: 89, 88. Step 6: Ex27: 93, 95; Ex28: 90, 100. Step 7: Assessment 7: 75, 80, 98, 100. (All numbers are verified against the book's printed answer key.) Final answer: 33, 36; 47, 43; 56, 54; 63, 66; 37, 47, 40, 62; 73, 77; 89, 88; 93, 95; 90, 100; 75, 80, 98, 100.

Question: Workbook — Addition (Exercises 29 to 32 and Assessment 8): (Ex29) Example 1 + 1 = 2; then: 2 + 1, 3 + 1, 4 + 1, 1 + 2. (Ex30) Example 1 + 4 = 5; then: 2 + 4, 3 + 4, 4 + 4, 1 + 5. (Ex31) Example 6 + 0 = 6; then: 7 + 0, 5 + 0, 8 + 0, 6 + 0. (Ex32 — vertical) Example 2 + 5 = 7; then: 3 + 5, 1 + 8, 2 + 6, 4 + 0. (Assessment 8) 1 + 2, 5 + 0, 3 + 3, 6 + 2, 2 + 7.

Answer: Step 1: Count forward from the first number as much as the second number. Step 2: Ex29: 2 + 1 = 3; 3 + 1 = 4; 4 + 1 = 5; 1 + 2 = 3. Step 3: Ex30: 2 + 4 = 6; 3 + 4 = 7; 4 + 4 = 8; 1 + 5 = 6. Step 4: Ex31: Adding zero keeps the number the same: 7, 5, 8, 6. Step 5: Ex32 (vertical): 3 + 5 = 8; 1 + 8 = 9; 2 + 6 = 8; 4 + 0 = 4. Step 6: Assessment 8: 1 + 2 = 3; 5 + 0 = 5; 3 + 3 = 6; 6 + 2 = 8; 2 + 7 = 9. Final answer: 3, 4, 5, 3; 6, 7,

8, 6; 7, 5, 8, 6; 8, 9, 8, 4; 3, 5, 6, 8, 9.

Question: Workbook — Subtraction (Exercises 33 to 36 and Assessment 9): (Ex33) Example 5 − 1 = 4; then: 3 − 2, 6 − 3, 5 − 2, 6 − 4. (Ex34) Example 8 − 3 = 5; then: 7 − 4, 8 − 6, 9 − 8, 7 − 5. (Ex35) Example 5 − 0 = 5; then: 1 − 0, 3 − 0, 7 − 0, 0 − 0. (Ex36 — vertical) Example 5 − 2 = 3; then: 5 − 3, 8 − 1, 6 − 2, 4 − 4. (Assessment 9) 5 − 1, 6 − 5, 2 − 0, 9 − 9, 8 − 1.

Answer: Step 1: Count backward from the first number as much as the second number. Step 2: Ex33: 3 − 2 = 1; 6 − 3 = 3; 5 − 2 = 3; 6 − 4 = 2. Step 3: Ex34: 7 − 4 = 3; 8 − 6 = 2; 9 − 8 = 1; 7 − 5 = 2. Step 4: Ex35: Subtracting zero keeps the number the same: 1, 3, 7, 0. Step 5: Ex36 (vertical): 5 − 3 = 2; 8 − 1 = 7; 6 − 2 = 4; 4 − 4 = 0. Step 6: Assessment 9: 5 − 1 = 4; 6 − 5 = 1; 2 − 0 = 2; 9 − 9 = 0; 8 − 1 = 7. Final answer: 1, 3, 3, 2; 3, 2, 1, 2; 1, 3, 7, 0; 2, 7, 4, 0; 4, 1, 2, 0, 7.

Question: Workbook — G1 Assessments 1, 2 and 3: (Assessment 1) Count the fingers and write: one, four, seven, nine, ten; two missing numbers in the 0–20 number line; bonds of 10: 10 = 2 + ___ and 10 = 7 + ___; tick the bigger number: from 7 and 6. (Assessment 2) Four block numbers. (Assessment 3) Add and subtract: 2 + 4, 7 + 0, 5 + 3, 8 + 1, 3 + 6; 4 − 3, 5 − 2, 7 − 0, 8 − 6, 9 − 7.

Answer: Step 1: Assessment 1: Fingers: 1, 4, 7, 9, 10. Step 2: Missing in the 0–20 number line: 6 and 17. Step 3: Bonds: 10 = 2 + 8; 10 = 7 + 3; bigger number: tick 7 (7 > 6). Step 4: Assessment 2: Count the tens blocks and ones: 39, 47, 86, 99 (as per key). Step 5: Assessment 3 — addition: 2 + 4 = 6; 7 + 0 = 7; 5 + 3 = 8; 8 + 1 = 9; 3 + 6 = 9. Step 6: Subtraction: 4 − 3 = 1; 5 − 2 = 3; 7 − 0 = 7; 8 − 6 = 2; 9 − 7 = 2. Final answer: Assessment 1: 1, 4, 7, 9, 10; 6, 17; 8; 3; ✓ 7. Assessment 2: 39, 47, 86, 99. Assessment 3: 6, 7, 8, 9, 9; 1, 3, 7, 2, 2.